When Phase Becomes a Clock - Complex Completion, Secondary Time, and the Search for Time-Bearing Worlds Across Domains

Complex Completion, Secondary Time, and the Search for Time-Bearing Worlds Across Domains

A Cross-Domain Extension of the R + iQ Framework after *"When Valuation Becomes a World, Part II"*

Abstract

Complex numbers are often introduced as an efficient way to store two real coordinates, represent rotation, or solve equations that have no real-valued roots. In mature scientific applications, however, their importance is rarely exhausted by compact notation. A complex state carries an intrinsic phase structure. That phase may sometimes provide a more natural internal ordering than ordinary calendar time.

This article develops a cross-domain research hypothesis abstracted from *When Valuation Becomes a World, Part II*. Let a protocol-bound system be represented by:

$$Z_P(t) = R_P(t) + iQ_P(t) = A_P(t) \exp[i\theta_P(t)]. \quad (0.1)$$

Here R_P is the structure currently admitted, expressed, verified, or committed under protocol P. Q_P is a retained, latent, compensatory, reactive, or unresolved conjugate structure. A_P is the total declared state magnitude, and θ_P is the orientation between admitted and retained structure.

At a static instant, Z = R + iQ contains no more numerical information than the ordered pair (R,Q). The stronger justification for complexification appears dynamically. The complex structure supplies a canonical phase generator, allowing uneven evolution in parent-world time t to be reparameterized as a more regular progression in θ.

Under locally stable amplitude:

$$dZ/d\theta \approx iZ. \quad (0.2)$$

A process may therefore advance irregularly in calendar time while traversing comparable internal phase distances. Two financial crises, biological recoveries, legal cases, software projects, scientific programmes, or institutional reforms may take radically different durations yet pass through structurally similar internal stages.

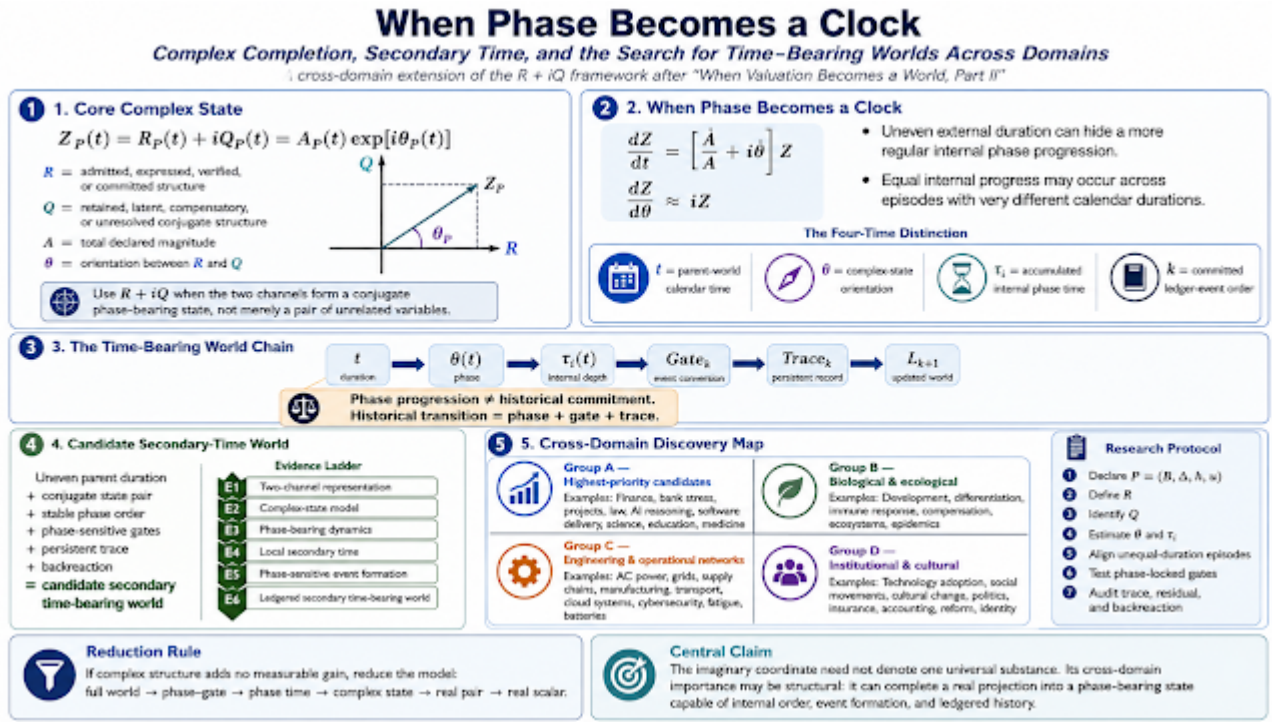
The article proposes that complex numbers should receive priority as a modelling language when a domain exhibits the following combination:

Uneven parent duration + conjugate state pair + stable phase order + phase-sensitive gates + persistent trace + backreaction = candidate secondary time-bearing world. (0.3)

A secondary time-bearing world is not established merely because a system oscillates or possesses two variables. The stronger claim requires phase to organize internal progression,

consequential transitions to occur within reproducible phase regions, gated events to enter persistent trace, and those traces to modify subsequent dynamics.

The article develops this proposition as a discovery programme across finance, AI, law, organizations, education, biology, ecology, engineering, infrastructure, science, and social institutions. It does not claim that these domains are physically quantum. It proposes that some may possess a protocol-bound complex geometry through which a projected real state acquires phase, internal ordering, event gates, and ledgered history.



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0. Reader's Guide — What This Article Claims and Does Not Claim

0.1 The central claim

The central claim is not that imaginary numbers secretly represent one universal hidden substance.

The claim is structural:

> Complexification may be the minimal mathematical move through which a real projection becomes a phase-bearing state capable of supporting an internal order.

A system may be observed in calendar time t while possessing another progression variable θ that better represents its internal movement toward consequential events.

The simplest relationship is:

$t \rightarrow \theta(t)$. (0.4)

But a full time-bearing world requires more than phase. It requires a transition from internal progression to committed history:

$t \to \theta(t) \to \tau_i(t) \to Gate_k \to Trace_k \to L_{k+1}$. (0.5)

Here:

- t is parent-world calendar duration;
- θ is complex-state orientation;
- τ_i is accumulated internal phase depth;
- $Gate_k$ decides whether a possible transition becomes an accepted event;
- $Trace_k$ records that event;
- L_{k+1} is the updated ledger that constrains future projection and action.

This distinction follows the broader world-formation principle that a world is not merely observed. It is declared, projected, gated, traced, audited, and revised under a bounded protocol.

0.2 What the article does not claim

This article does not claim:

Every pair of real variables should be written as a complex number. (0.6)

Every oscillation generates internal time. (0.7)

Every latent variable belongs on an imaginary axis. (0.8)

Every phase coordinate produces a secondary universe. (0.9)

Social systems are physically quantum. (0.10)

Biological compensation is literally reactive electrical power. (0.11)

Legal residual is literally quantum uncertainty. (0.12)

The imaginary unit i has one universal material interpretation. (0.13)

The proposed cross-domain transfer is functional and protocol-bound. It asks whether corresponding domains solve similar world-formation problems through admitted state, retained conjugate structure, phase progression, event gates, trace, residual, and revision.

The legitimate form is:

Domain A $\cong_P$ Domain B only when the role map preserves function, gate behaviour, trace effect, residual structure, and relevant invariants under declared protocol P. (0.14)

This is stronger than loose analogy but weaker than literal identity. The protocol-bound approach treats interfaces as world-generating structures and isomorphisms as recurring functional anatomy, while insisting that the mapped substances remain different.

0.3 Five evidence levels

The article distinguishes five levels of claim.

Level 1 — Descriptive pairing

The domain contains two useful variables R and Q.

No complex interpretation is yet required.

Level 2 — Complex completion

R and Q form a meaningful conjugate state:

$Z = R + iQ.$ (0.15)

The magnitude and phase have defensible interpretations.

Level 3 — Dynamical phase

Phase θ simplifies the description of system evolution.

The phase-indexed model is more stable, transferable, or predictive than the calendar-time model.

Level 4 — Secondary internal time

Unequally paced episodes become structurally comparable when indexed by θ or accumulated phase τ_i.

Level 5 — Time-bearing world

Phase-sensitive gates write persistent trace, alter future admissibility, and backreact on the parent environment.

Only Level 5 supports the strongest expression:

> The domain contains a secondary time-bearing world.

Part I — Why Complex Numbers May Generate Internal Time

1. The Starting Question: Why R + iQ Rather Than (R,Q)?

1.1 Static equivalence

At a single instant, the ordered pair:

$(R,Q) \in \mathbb{R}^2$ (1.1)

and the complex number:

$Z = R + iQ \in \mathbb{C}$ (1.2)

contain the same two scalar values.

A critic may therefore ask:

> Why introduce an imaginary axis? Why not retain two ordinary real variables?

If the task is merely to record two measurements, the criticism is correct. Complex notation by itself adds no empirical information.

A temperature and a pressure do not become a meaningful complex state merely because they can be written:

$$Z = \text{Temperature} + i\,\text{Pressure}. \quad (1.3)$$

Two variables must possess an additional structure before complexification becomes scientifically useful.

1.2 A complex state carries a canonical generator

The first important difference is that a complex state comes equipped with multiplication by i:

$$JZ = iZ. \quad (1.4)$$

This operator satisfies:

$$J^2 = -I. \quad (1.5)$$

Applied to $Z = R + iQ$:

$$i(R + iQ) = -Q + iR. \quad (1.6)$$

Therefore:

$$R \rightarrow Q. \quad (1.7)$$

$$Q \rightarrow -R. \quad (1.8)$$

This is not merely a relabelling of two coordinates. It specifies a quarter-turn relation between them.

In a bare ordered pair, no unique transformation is privileged. One may define any two-dimensional matrix to rotate, stretch, shear, reflect, or mix the coordinates. The pair alone does not say which transformation is intrinsic.

A complex structure adds a distinguished orientation-preserving operator J whose square is minus the identity.

The substantive question is therefore not:

> Can these two quantities be placed on two axes?

It is:

> Does the domain possess a stable transformation for which multiplication by i, or a generalized complex generator, captures a real conversion between the two channels?

If no such transformation exists, the complex representation may remain ornamental.

1.3 R and Q must be conjugate, not merely different

The proposed general interpretation is:

R_P = admitted, expressed, verified, or committed structure under protocol P. (1.9)

Q_P = retained, latent, compensatory, reactive, or unresolved conjugate structure under protocol P. (1.10)

The word conjugate is essential.

Q is not simply a second variable. It must be dynamically related to R.

A candidate Q should satisfy at least four conditions:

1. Q is not already contained in R.
2. Q affects future movement in R.
3. Some part of Q can convert into, support, oppose, or regulate R.
4. The R–Q relation remains sufficiently stable under repeated episodes or admissible frame changes.

Examples may include:

- admitted value and retained valuation pressure;
- observable physiological function and compensatory regulatory effort;
- completed work and unresolved dependency pressure;
- accepted legal fact and appeal-bearing residual;
- published scientific structure and anomaly pressure;
- committed AI answer and unresolved alternative hypotheses.

These examples do not share one substance. Their possible similarity lies in a functional relation between admitted structure and a retained conjugate channel.

1.4 The total state and its orientation

If R and Q form a meaningful complex state, one may write:

$$Z(t) = R(t) + iQ(t). \quad (1.11)$$

The polar form is:

$$Z(t) = A(t) \exp[i\theta(t)]. \quad (1.12)$$

where:

$$A^2(t) = R^2(t) + Q^2(t). \quad (1.13)$$

$$R(t) = A(t) \cos \theta(t). \quad (1.14)$$

$$Q(t) = A(t) \sin \theta(t). \quad (1.15)$$

The magnitude A represents the total declared state under the chosen protocol and norm.

The phase θ represents how that total state is divided between the admitted and retained channels.

The phase can be recovered through:

$$\theta(t) = \operatorname{atan2}[Q(t),R(t)]. \quad (1.16)$$

This already introduces a potentially powerful distinction.

Two systems may display the same real-side value R while possessing different Q and therefore different θ.

For example:

- two banks may report the same liquidity ratio but carry different contingent funding pressure;
- two patients may show the same measured function but require different compensatory effort;
- two students may produce the same test score but possess different latent understanding and misconception structures;
- two software projects may report the same completion percentage but carry different dependency and defect pressure.

Therefore:

Same R does not imply same Z. (1.17)

Same observable projection does not imply same internal phase. (1.18)

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2. From Complex Completion to Phase Dynamics

2.1 Calendar-time evolution

Let:

$$Z(t) = A(t) \exp[i\theta(t)]. \quad (2.1)$$

Differentiating with respect to parent-world time t gives:

$$dZ/dt = [\dot{A}(t)/A(t) + i\dot{\theta}(t)]Z(t). \quad (2.2)$$

This equation separates two different forms of change:

- radial change through $\dot{A}/A$;
- angular change through $\dot{\theta}$.

A system may therefore change because:

1. its total magnitude A is growing or shrinking;

2. its state is rotating between R and Q;

3. both processes are occurring simultaneously;

4. an external forcing or residual term is disturbing the ideal complex motion.

A more general model is:

$dZ/dt = [g_A(t) + i\omega(t)]Z(t) + \varepsilon(t)$. (2.3)

where:

$g_A(t) = \dot{A}(t)/A(t)$. (2.4)

$\omega(t) = \theta(t)$. (2.5)

$\varepsilon(t)$ = unresolved forcing, model error, or non-complex residual. (2.6)

This distinction matters because visible change in R does not automatically mean growth in the total system.

R may rise because A increased.

R may also rise because θ rotated an existing Q component into the admitted axis.

Conversely, R may remain stable while A and Q change substantially.

Therefore:

Observed real-axis change = amplitude change + phase conversion + residual effect. (2.7)

This is one of the central analytical advantages of a phase-bearing representation.

2.2 Uneven rotation in parent time

The angular velocity:

$\omega(t) = d\theta/dt$ (2.8)

need not be constant.

A process may:

- advance rapidly;
- slow down;
- remain temporarily stalled;
- reverse direction;
- jump following an external intervention;
- accelerate near a transition gate.

This means that equal calendar intervals need not represent equal internal progress.

Suppose two episodes traverse the same phase distance:

$\Delta\theta_1 = \Delta\theta_2$. (2.9)

But they require different calendar durations:

$\Delta t_1 \neq \Delta t_2$. (2.10)

The episodes are unequal in parent-world time but equal in phase distance.

This suggests a secondary ordering:

> Internal progress may be measured by how far the state has rotated, not by how long the external clock has run.

A market may traverse a credit-stress phase in three years during one cycle and three weeks during another.

A patient may cross the same recovery phase in two months or one year.

A legal case may reach the same procedural depth after twenty hearings or three negotiated steps.

A software project may spend six months near one phase and cross several release-readiness phases in a single week.

Calendar time records duration.

Phase may record structural progression.

2.3 Phase reparameterization

When θ is locally monotonic, the chain rule gives:

$dZ/d\theta = (dZ/dt)/(d\theta/dt)$. (2.11)

Substituting the general law:

$dZ/d\theta = [g_A(t)/\omega(t) + i]Z + \varepsilon(t)/\omega(t)$. (2.12)

If amplitude changes slowly relative to phase and residual forcing is small:

$g_A/\omega \approx 0$. (2.13)

$\varepsilon/\omega \approx 0$. (2.14)

then:

$dZ/d\theta \approx iZ$. (2.15)

The uneven calendar-time speed $\omega(t)$ disappears from the internal phase law.

The corresponding channel equations are:

$dR/d\theta \approx -Q$. (2.16)

$dQ/d\theta \approx R$. (2.17)

This is the decisive result.

The system may be highly irregular in t while remaining approximately regular in θ.

Therefore, the stronger answer to the question “Why use R + iQ rather than (R,Q)?” is:

> Because a complex representation can expose a phase coordinate under which irregular external evolution becomes a simpler internal law.

The ordered pair records two channels.

The complex state may generate an internal clock.

2.4 The meaning of even internal time

The phrase “even iTime” requires precision.

It does not mean the parent-world state moves at constant calendar speed.

It means equal intervals of internal time correspond to equal phase distances.

For a locally monotonic phase, define:

$\tau_i(t) = \text{unwrap}[\theta(t)]$. (2.18)

Then:

$\Delta\tau_i = \text{constant}$ (2.19)

represents equal phase progress even if:

Δt varies. (2.20)

An alternative normalized definition is:

$\tau_i(t) = \text{unwrap}[\theta(t)]/\Omega^*$. (2.21)

where Ω^* is a chosen reference scale.

The internal world can then treat equal increments of τ_i as equal steps, while the parent world sees those steps taking unequal durations.

This produces the conceptual split:

Parent-world time asks: How long did it take? (2.22)

Secondary phase time asks: How far did the internal state progress? (2.23)

The distinction resembles the use of developmental stage rather than age, procedural stage rather than elapsed days, readiness rather than project duration, or risk phase rather than calendar date.

But complex secondary time adds a stronger demand: those stages should arise from a conjugate R–Q state and a reproducible phase geometry, not merely from manually assigned categories.

2.5 Phase reversal and branch structure

A global phase clock cannot be assumed automatically.

If:

$\dot{\theta}(t)$ changes sign, (2.24)

then the same phase angle may be visited more than once.

A system may move forward toward a gate, retreat, and later return to the same orientation with a different history.

Therefore:

Same θ does not imply same historical state. (2.25)

One solution is to use unwrapped phase with branch labels:

$\tau_\theta = (b,\theta)$. (2.26)

where b identifies the relevant phase branch or traversal.

Another solution is cumulative phase distance:

$\tau_i(t) = \int_0^t |\dot{\theta}(s)|\, ds$. (2.27)

This quantity remains monotonic through reversals, but it loses direction unless direction is stored separately.

A complete internal-time state may therefore require:

InternalTimeState = $(\tau_i, \operatorname{sgn} \dot{\theta}, \text{branch}, \text{ledger})$. (2.28)

This reveals why phase alone is insufficient.

Two systems can occupy the same angle but differ in:

- direction;
- prior loops;
- accumulated residual;
- already crossed gates;
- ledgered commitments.

The secondary world needs memory.

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3. Four Distinct Forms of Time

The discussion requires a careful separation among calendar time, phase, accumulated internal time, and ledger time.

3.1 Calendar time t

Calendar time measures duration in the parent world.

It orders external observations and supplies the independent variable used by ordinary time-series models.

Calendar time answers:

- How many seconds passed?
- How many days elapsed?
- How old is the system?
- How long did the episode last?

But it does not necessarily reveal how far the system progressed internally.

3.2 Phase θ

Phase measures orientation in the complex state:

$\theta = \arg Z.$ (3.1)

It indicates the current division between R and Q.

Phase may represent:

- valuation orientation;
- stress orientation;
- readiness orientation;
- developmental orientation;
- evidential consolidation;
- procedural depth;
- institutional commitment;
- regulatory compensation.

Phase is cyclical or locally charted. It does not by itself guarantee monotonic time.

3.3 Secondary phase time τ_i

Secondary phase time accumulates phase progression.

For monotonic motion:

$\tau_i(t) = \text{unwrap}[\theta(t)].$ (3.2)

For reversal-sensitive distance:

$\tau_i(t) = \int_0^t |d\theta/ds|\, ds.$ (3.3)

This coordinate makes differently paced episodes comparable by internal progress rather than

external duration.

It may answer:

- How far has the process moved toward a transition?
- How much internal phase has been traversed?
- Are two episodes structurally at the same stage despite different ages?
- Does the same event tend to occur at the same phase?

3.4 Ledger time k

Ledger time advances only when a gate converts a possible transition into accepted trace.

Let:

$e_k = \text{Gate_P}[Z(\tau_i)]$. (3.4)

Then:

$L_{k+1} = \text{UpdateLedger_P}(L_k, e_k)$. (3.5)

The index k counts committed events rather than continuous phase increments.

Examples include:

- trade execution;
- margin breach;
- legal judgment;
- software release;
- diagnosis;
- scientific publication;
- policy adoption;
- memory write;
- institutional reform.

Ledger time answers:

- What has officially happened?
- What is now binding?
- What can later observers inspect?
- What has changed future admissibility?

Thus:

Phase progression ≠ historical commitment. (3.6)

Historical commitment = phase-sensitive gate + persistent trace. (3.7)

3.5 The four-time table

Coordinate	Primary function	Main question
t	Parent-world duration	How long has the process run?
θ	Complex-state orientation	Where is the state between R and Q?
τ_i	Accumulated internal phase depth	How far has the internal process progressed?
k	Committed ledger-event order	What consequential events have become history?

The complete chain is:

$t \rightarrow Z(t) \rightarrow \theta(t) \rightarrow \tau_i(t) \rightarrow Gate_k \rightarrow Trace_k \rightarrow L_{k+1}$. (3.8)

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4. What Counts as a Secondary Time-Bearing World?

4.1 A graded ladder

The phrase “secondary universe” should not be used at the first appearance of a phase variable.

A disciplined framework requires several levels.

Level 1 — Two-channel description

The system has:

R(t), Q(t). (4.1)

The variables may be useful but unrelated.

Level 2 — Complex completion

The system supports:

$Z(t) = R(t) + iQ(t)$. (4.2)

The complex magnitude, phase, and i-generated transformation possess operational interpretations.

Level 3 — Phase-bearing dynamics

The phase:

$\theta(t) = \text{atan2}[Q(t),R(t)]$ (4.3)

simplifies prediction, comparison, or control.

Level 4 — Secondary ordering

Differently paced episodes align better in θ or τ_i than in t.

Level 5 — Phase-sensitive event formation

Consequential transitions occur within stable phase regions:

$\theta_gate \in [\theta_-, \theta_+]$. (4.4)

or:

$\tau_i,gate \in [\tau_-, \tau_+]$. (4.5)

Level 6 — Ledgered secondary world

Gated events:

- become persistent trace;
- condition future observation;
- constrain future action;
- alter the state geometry;
- backreact on the parent world.

Only at this level does the system become a strong candidate secondary time-bearing world.

4.2 Working definition

A candidate secondary time-bearing world may be defined as follows:

> A protocol-bounded domain in which a conjugate complex state generates a meaningful phase order, phase progression provides a more stable internal coordinate than parent-world duration, consequential events are phase-sensitive, and gated traces modify subsequent dynamics.

In compact form:

SecondaryWorld_P = ComplexState_P + PhaseOrder_P + Gate_P + Trace_P + Backreaction_P. (4.6)

The protocol subscript matters.

The same underlying process may generate different complex states under different choices of:

- boundary;
- measurement rule;
- horizon;
- admissible intervention;

- baseline;
- feature map.

Therefore:

$P = (B, \Delta, h, u)$. (4.7)

$World_P = (X, q, \varphi, P)$. (4.8)

A secondary time claim is never made about “the domain as such.” It is made under a declared protocol.

4.3 The central diagnostic formula

The strongest candidate structure is:

Uneven parent duration

- conjugate state pair
- stable phase order
- phase-locked gates
- ledgered consequence
- parent-world backreaction
  = secondary time-bearing world candidate. (4.9)

Each term is necessary.

Without uneven parent duration

There may be no practical reason to distinguish internal time from calendar time.

Without a conjugate state pair

The phase may be an arbitrary index rather than a complex-state orientation.

Without stable phase order

The model may not support a usable internal clock.

Without phase-sensitive gates

Phase may describe motion without organizing consequential events.

Without trace

The secondary process lacks history.

Without backreaction

The secondary world may remain a passive representation rather than an operational world.

4.4 The role of residual

No phase model closes the entire parent system.

Let:

Residual_P = structure not absorbed by Z_P under the declared protocol. (4.10)

A mature complex model must preserve:

- unexplained forcing;
- branch ambiguity;
- measurement error;
- omitted degrees of freedom;
- failed phase alignment;
- gate exceptions;
- cross-frame inconsistency.

Thus:

MatureComplexClosure_P = ComplexState_P + ResidualRegister_P + RevisionPath_P. (4.11)

The model should not absorb every mismatch into Q.

If Q becomes a universal container for everything unexplained, the theory becomes unfalsifiable.

The proper relationship is:

Q = declared conjugate structure. (4.12)

Residual = what the declared complex model still fails to absorb. (4.13)

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5. A General Translation Grammar

Before examining candidate domains, the framework requires a shared vocabulary.

Element	General operational role
R	Admitted, expressed, verified, observable, or committed structure
Q	Retained, latent, reactive, compensatory, or unresolved conjugate structure
A	Total declared state magnitude or capacity
θ	Orientation between admitted and retained structure
τ_i	Accumulated internal phase depth
t	Parent-world calendar duration
Gate	Threshold converting possible state into consequential event
Trace	Persistent record that affects later projection or action
k	Committed ledger-event order
Residual	Remainder not absorbed by the current complex closure

Element	General operational role
Backreaction	Effect of a secondary-world output on the parent environment
P	Declared boundary, observation rule, horizon, and intervention family

The general runtime is:

$$\Sigma_0 \to \text{Declare_P} \to \text{Project_P} \to Z_P = R_P + iQ_P \to \theta_P \to \tau_{i,P} \to \text{Gate_P} \to \text{Trace_P} + \text{Residual_P} \to \text{InvarianceTest_P} \to \text{Revise_P} \to \Sigma'. \quad (5.1)$$

This chain combines two ideas.

The first is complex completion: a real projection and a conjugate retained structure generate phase.

The second is observer-compatible world formation: projection becomes history only through gate, trace, residual preservation, invariance, and revision.

The complex state may supply the internal clock.

The world-formation architecture supplies the conditions under which that clock becomes operationally consequential.

5.1 The complex-first criterion

A domain should receive priority for complex modelling when it satisfies most of the following conditions:

1. Calendar duration poorly represents internal progress.
2. Repeated episodes traverse comparable stages at different speeds.
3. A defensible real-side state R can be measured.
4. A distinct conjugate channel Q can be independently estimated.
5. R and Q display systematic conversion, compensation, opposition, or phase coupling.
6. Phase θ simplifies dynamics or aligns episodes.
7. Major gates occur within stable phase regions.
8. Gated events enter persistent trace.
9. Trace alters future behaviour or admissibility.
10. The complex model outperforms a flexible two-real-variable alternative.

This gives:

$$\text{ComplexFirst_P} \Leftrightarrow \text{Conjugacy_P} \wedge \text{PhaseUtility_P} \wedge \text{GateRelevance_P} \wedge \text{TraceEffect_P}. \quad (5.2)$$

5.2 When the ordered pair should remain primary

A real pair should remain the preferred representation when:

- R and Q are merely correlated;
- no meaningful magnitude A exists;
- phase depends arbitrarily on measurement scaling;
- no stable rotational or phase-coupled relation appears;
- phase does not simplify the dynamics;
- event gates do not align in phase;
- a flexible real-state model performs equally well or better.

In those cases:

Use (R,Q), not R + iQ. (5.3)

Complex numbers should be prioritized where they generate explanatory structure, not where they merely decorate a graph.

Next installment: Part III — Candidate Domain Discovery Maps, beginning with the complete Table A for highest-priority domains and Table B for biological and ecological candidates.

Part III — Candidate Domain Discovery Maps

6. Domain Group A — Highest-Priority Candidates

6.1 Why these domains should be tested first

The highest-priority candidates are not necessarily the domains in which complex secondary time is most philosophically interesting. They are the domains in which the hypothesis can be tested most clearly.

A strong candidate should provide:

- repeated episodes;
- large differences in calendar duration;
- measurable state variables;
- identifiable transition gates;
- persistent event records;
- observable effects of earlier commitments on later dynamics.

These conditions make it possible to distinguish a genuine phase-bearing model from retrospective storytelling.

The research question is not merely:

> Can an internal stage variable be assigned?

It is:

> Can independently measured R and Q generate a phase θ that aligns differently paced episodes, predicts consequential gates, and remains useful outside the data from which it was estimated?

The target signature is:

$$\mathrm{Var}[\text{Trajectory} \mid \theta] < \mathrm{Var}[\text{Trajectory} \mid t]. \quad (6.1)$$

That is, trajectories should become more comparable when indexed by phase rather than calendar time.

The stronger gate test is:

$$\Pr(\text{Gate} = 1 \mid \theta) > \Pr(\text{Gate} = 1 \mid t, \text{ordinary state controls}). \quad (6.2)$$

The strongest operational test is:

$$\text{Outcome}[\text{Intervention}(\theta)] > \text{Outcome}[\text{Intervention}(t)]. \quad (6.3)$$

A phase-aware intervention should outperform one based only on elapsed duration or a conventional single-axis indicator.

6.2 Table A — Highest-Priority Candidate Domains

Domain	Why calendar time is inadequate	Candidate R	Candidate Q	Candidate θ or τ_i	Gate and ledger event
Derivatives and credit markets	Credit deterioration, volatility transition, and liquidity collapse may unfold over years in one episode and days in another while passing through comparable risk configurations	Executable price, booked value, recognized exposure, settled cash flow	Implied risk pressure, hedge demand, liquidity stress, unrecognized tail exposure, collateral pressure	Valuation orientation or credit-stress phase	Margin call, downgrade, exercise, restructuring, default, clearin settlement
Bank balance-sheet stress	Reported solvency can remain stable while internal funding pressure accelerates; crisis speed depends on withdrawal,	Reported liquidity, recognized capital, available funding, settled asset value	Contingent liquidity demand, collateral immobility, deposit-flight pressure, hidden leverage, off-balance-sheet	Funding-to-liquidation phase	Collateral breach, emergency borrowing, supervisory intervention, resolution, insolvency

Domain	Why calendar time is inadequate	Candidate R	Candidate Q	Candidate θ or τ_i	Gate and ledger event
	collateral, and confidence dynamics		obligation		
Project and programme delivery	Projects can appear unchanged for months and then pass several milestones rapidly; elapsed time often measures delay rather than progress	Verified deliverables, accepted functionality, completed dependency chain	Rework burden, unresolved integration pressure, hidden scope, blocked dependencies, approval debt	Delivery-readiness or integration phase	Design approva integration pas release authorization, customer acceptance, formal closure
Organizational transformation	Formal announcements occur early, while actual operating change may take months or years; equal calendar age does not imply equal institutional adoption	Officially adopted roles, policies, workflows, budgeted practices	Resistance, shadow processes, legacy incentives, political debt, unowned work, cultural residual	Institutional commitment or internalization phase	Budget commitment, policy enforcement, workflow adoption, structural reorganization, audit confirmation
Legal proceedings	Cases with similar procedural depth may last radically different periods due to scheduling, negotiation, jurisdiction, and evidence complexity	Admitted facts, accepted claims, procedural status, enforceable judgment	Excluded evidence, unresolved harm, appeal pressure, doctrinal ambiguity, enforcement risk	Procedural-admissibility phase	Evidence admission, ruling, judgment, appeal, settlement, enforcement
AI reasoning and agent runtimes	One task may require five operations and another fifty while passing through comparable uncertainty, verification, and	Selected hypothesis, verified proposition, approved action, published answer	Alternative hypotheses, conflicting evidence, unverified tool output, uncertainty, unresolved policy conflict	Inference-commitment phase	Tool authorization, verifier pass, answer publication, memory write, human handof

Domain	Why calendar time is inadequate	Candidate R	Candidate Q	Candidate θ or τ_i	Gate and ledger event
	commitment stages				
Software delivery and DevOps	Calendar age of a branch or release says little about readiness; progress may stall until a dependency resolves and then accelerate	Tested, integrated, deployable functionality	Open defects, security exposure, dependency pressure, rollback risk, technical debt	Release-readiness phase	Merge, build pass, deployment, rollback, incident declaration, incident closur
Scientific research programmes	A theory may remain unresolved for decades and then advance rapidly after a new instrument or result; publication chronology poorly measures evidential maturity	Reproduced findings, accepted model structure, verified predictions	Anomalies, unexplained variance, failed replication, unresolved alternatives, instrumental uncertainty	Evidence-consolidation or theory-readiness phase	Experiment confirmation, replication, publication, consensus shift theory revision
Education and skill acquisition	Equal instruction hours do not produce equal developmental progress; plateaus and sudden integration are common	Demonstrated transferable competence, stable performance, independent problem solving	Misconception pressure, latent understanding, unused capacity, cognitive overload, dependence on prompts	Learning-integration phase	Successful transfer, independent solution, mastery decision, progression to a new level
Clinical disease progression	Patients may reach the same clinical stage over very different periods; visible function can	Observable physiological function, accepted diagnosis, measured performance	Compensatory effort, inflammatory pressure, latent pathology, reserve depletion,	Disease-compensation phase	Diagnosis, decompensatio hospitalization, treatment escalation, remission

Domain	Why calendar time is inadequate	Candidate R	Candidate Q	Candidate θ or τ_i	Gate and ledger event
	remain stable while compensation is failing		regulatory strain		
Recovery and rehabilitation	Recovery contains plateaus, regressions, and sudden gains; elapsed weeks often fail to represent readiness	Stable demonstrated function, tolerated activity, verified movement capacity	Latent strength, neural adaptation, fatigue, pain avoidance, reinjury pressure	Recovery-readiness phase	Return to work return to sport, discharge, relapse, treatment transition
Neural decision formation	Reaction time varies across trials and subjects although internal evidence may pass through comparable commitment stages	Selected percept, motor plan, reportable decision	Competing representations, uncertainty, unresolved prediction error, inhibitory pressure	Evidence-to-commitment phase	Conscious report, action initiation, memory encoding

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6.3 Finance as the initial worked example

Finance remains the most developed example because the distinction between projected value and retained pressure is already embedded in mature practice.

A financial system distinguishes among:

- quoted value;
- executable value;
- model value;
- funded value;
- collateral value;
- accounting value;
- regulatory value;
- liquidation value.

These values do not merely offer different opinions about one static object. They belong to

different local frames and become consequential through different gates.

A model valuation becomes a financial event only when it affects:

- execution;
- margin;
- collateral;
- capital allocation;
- accounting recognition;
- clearing;
- settlement;
- legal obligation.

The secondary valuation world therefore contains its own internal progression and event history.

A simplified chain is:

Market conditions → valuation state → phase rotation → trade or risk gate → ledger update → market backreaction. (6.4)

The backreaction may then change:

- price;
- liquidity;
- volatility;
- collateral availability;
- funding cost;
- counterparty behaviour.

This makes finance more than a descriptive test case. It illustrates how a secondary model world can alter the primary system that originally generated it.

6.4 Bank stress as a particularly strong test

Banking crises provide a clear example of why calendar time can be misleading.

A bank may report stable capital and liquidity for an extended period. During that time, however:

- collateral mobility may deteriorate;
- deposit concentration may rise;
- contingent funding obligations may accumulate;
- market confidence may weaken;

- legally transferable liquidity may shrink.

The visible real-side state may remain approximately constant:

$dR/dt \approx 0$. (6.5)

while retained pressure grows:

$dQ/dt > 0$. (6.6)

The phase then rotates even when the reported state appears stable:

$d\theta/dt > 0$. (6.7)

A crisis occurs when a gate is crossed:

$\theta \geq \theta_{liquidity}$. (6.8)

The gate may be triggered by:

- a collateral call;
- a large withdrawal;
- a ratings action;
- a public disclosure;
- a failed asset sale.

The important hypothesis is not that every bank crisis occurs at one universal angle. It is that, under a carefully declared protocol, crisis episodes may occupy comparable phase regions even when their calendar speeds differ.

A practical model might begin with:

R_{bank} = normalized admitted liquidity and capital capacity. (6.9)

Q_{bank} = normalized contingent funding and collateral pressure. (6.10)

$Z_{bank} = R_{bank} + iQ_{bank}$. (6.11)

Researchers could then compare:

Hazard(default | t, conventional ratios) (6.12)

against:

Hazard(default | θ_{bank}, A_{bank}, conventional ratios). (6.13)

If θ adds no out-of-sample value, the complex hypothesis should be rejected or revised.

6.5 Projects and software as accessible experimental environments

Project delivery and software engineering may offer easier initial research environments than finance or medicine.

They provide:

- detailed event logs;
- explicit gates;
- repeatable workflows;
- version histories;
- issue records;
- dependency graphs;
- build and release outcomes.

A simple project state might be:

R_project = verified integrated deliverables. (6.14)

Q_project = unresolved dependency, defect, and rework pressure. (6.15)

Z_project = R_project + iQ_project. (6.16)

Two projects may both report 80 per cent completion while occupying very different states.

Project A may have high verified integration and low unresolved pressure.

Project B may have extensive nominal completion but severe integration and rework debt.

Therefore:

R_A ≈ R_B (6.17)

does not imply:

Z_A ≈ Z_B. (6.18)

The release gate may depend more strongly on θ than on reported completion.

A testable proposition is:

Pr(successful release | θ_project) > Pr(successful release | reported completion, elapsed time). (6.19)

Because software repositories already preserve ordered trace, this domain may allow relatively direct reconstruction of phase, gate, and ledger relationships.

6.6 AI reasoning as a secondary-world candidate

AI reasoning provides a distinctive case because its operational time is already separated from human calendar time.

An AI agent may pass through:

- prompt interpretation;
- context selection;

- retrieval;
- tool use;
- hypothesis generation;
- verification;
- output gating;
- memory update.

The number of tokens or tool calls is not necessarily the relevant internal time.

A task completed in ten operations may be internally less mature than one completed in five, depending on the unresolved residual.

A candidate state is:

R_AI = verified support incorporated into the current answer or action. (6.20)

Q_AI = unresolved alternatives, unsupported assumptions, conflicting evidence, and unverified outputs. (6.21)

Z_AI = R_AI + iQ_AI. (6.22)

A premature answer occurs when the publication gate opens at an unstable phase:

Gate_publish[Z_AI] = 1 before θ enters the verified commitment band. (6.23)

A mature AI runtime would distinguish:

- phase progression;
- answer publication;
- persistent memory;
- residual carried forward.

Thus:

Reasoning progress ≠ answer commitment. (6.24)

Answer commitment ≠ memory write. (6.25)

Memory write ≠ admissible self-revision. (6.26)

This makes AI a promising domain for phase–gate–ledger research.

6.7 Law as event-bearing procedural time

Legal systems already distinguish calendar time from procedural time.

The internal legal world is ordered by events such as:

- filing;

- service;
- evidence admission;
- hearing;
- ruling;
- judgment;
- appeal;
- enforcement.

Two cases can have equal calendar age but radically different procedural depth.

A candidate legal state is:

R_law = legally admitted and currently enforceable structure. (6.27)

Q_law = unresolved evidential, doctrinal, enforcement, and appeal pressure. (6.28)

Z_law = R_law + iQ_law. (6.29)

The legal phase would not measure moral truth directly. It would measure orientation inside a declared legal protocol.

A judgment gate may transform contested possibility into official trace:

Dispute field → admissibility projection → judgment gate → legal ledger. (6.30)

But residual remains:

Mature legal closure = judgment + appeal residual + enforcement uncertainty + revision path. (6.31)

This distinguishes official closure from total resolution.

6.8 Science and education as observer-forming domains

Science and education are especially important because their gates do not merely classify external objects. They also form future observers.

In science, a result becomes historical through:

- measurement;
- replication;
- publication;
- citation;
- theory integration.

In education, understanding becomes stable through:

- exercise;
- error;
- feedback;
- transfer;
- independent reconstruction.

A scientific phase may order movement from conjecture to robust evidence.

An educational phase may order movement from exposure to transferable competence.

In both cases, calendar duration is weak because internal transformation is nonlinear.

The strongest test is transfer.

For education:

TransferSuccess = Gate_learning[Z_student]. (6.32)

For science:

ReplicationSuccess = Gate_science[Z_theory]. (6.33)

A phase model becomes valuable only if it predicts transfer or replication better than time spent, publication count, or nominal progress.

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7. Domain Group B — Biological and Ecological Candidates

7.1 Why biology may contain strong secondary-time structure

Biology already uses many internal time concepts:

- developmental stage;
- cell-cycle phase;
- circadian phase;
- immune activation stage;
- disease stage;
- physiological age;
- ecological succession stage.

But not every biological stage implies a complex secondary time.

The stronger hypothesis requires a conjugate relation between:

- expressed or maintained function;

- retained developmental, compensatory, or regulatory structure.

A biologically plausible complex model would distinguish visible function from the hidden work required to sustain it.

This gives a central proposition:

> A stable observable state may conceal rapid internal phase progression when compensatory structure is changing.

In symbols:

$dR/dt \approx 0$ while $dQ/dt \neq 0$. (7.1)

Consequently:

$d\theta/dt \neq 0$. (7.2)

Visible stability may therefore coexist with internal movement toward either resilience or failure.

7.2 Table B — Biological and Ecological Candidate Domains

Domain	Parent-time irregularity	Candidate R	Candidate Q	Candidate secondary time	Gate or trace	C
Embryonic development	Homologous developmental events occur at different speeds across species, individuals, and environments	Expressed differentiated structure and established morphology	Remaining developmental potential, morphogen tension, epigenetic readiness, unresolved patterning	Differentiation or morphogenesis phase	Cell-fate commitment, tissue formation, organ-pattern transition	C
Cell differentiation	Cells may remain poised for long periods and then commit rapidly after a regulatory threshold	Expressed lineage programme, stable phenotype markers	Competing lineage potential, chromatin readiness, regulatory conflict, latent response capacity	Fate-commitment phase	Irreversible regulatory switch, lineage stabilization, loss of alternative potential	C

Domain	Parent-time irregularity	Candidate R	Candidate Q	Candidate secondary time	Gate or trace	
Immune response	Immune activation, escalation, resolution, and memory formation occur at highly variable speeds	Measured protective response, active antibodies, effective clearance	Latent clonal capacity, unresolved antigenic load, inflammatory pressure, regulatory counter-response	Immune activation–resolution phase	Seroconversion, pathogen clearance, cytokine escalation, memory-cell formation	A r a s p t p e c p c f p c
Homeostasis and physiological compensation	Observable variables may remain normal for years while regulatory effort increases and reserve declines	Maintained physiological function and normal measured output	Compensatory drive, reserve depletion, autonomic or hormonal effort, hidden stress	Compensation-depth phase	Decompensation, acute crisis, adaptive reset, treatment threshold	I c r s t h r p f
Metabolic adaptation	Weight, glucose, or energy output may remain stable while hormonal and substrate regulation changes internally	Observable metabolic performance and maintained output	Insulin resistance, hormonal drive, substrate imbalance, energetic debt	Metabolic adaptation phase	New steady state, metabolic syndrome threshold, recovery, treatment transition	S r p r b v a c r
Ecological succession	Similar ecosystem transitions unfold over different durations depending on climate, disturbance, migration, and resource conditions	Visible species composition, biomass, canopy or trophic structure	Seed bank, resilience reserve, nutrient debt, disturbance pressure, latent competitors	Successional or resilience phase	Regime shift, canopy closure, trophic reorganization, collapse, recovery	A e p c t t c s r
Population collapse and recovery	Population counts may remain stable while reproductive capacity and	Current observed population and realized recruitment	Reproductive debt, genetic pressure, habitat degradation, predation and	Viability phase	Recruitment failure, collapse, recolonization, stable recovery	U r a i e t

Domain	Parent-time irregularity	Candidate R	Candidate Q	Candidate secondary time	Gate or trace	
	habitat support deteriorate		resource stress			p a v
Epidemics	Outbreaks with similar structure unfold at radically different speeds due to transmission, behaviour, immunity, and intervention	Detected infections, recoveries, recognized immunity, available healthcare capacity	Latent infections, susceptibility, mobility pressure, behavioural response, reporting delay	Epidemic propagation–resolution phase	Outbreak declaration, peak, capacity overload, control, endemic transition	A c p t v a g r
Evolutionary transitions	Adaptive innovations, fixation, and speciation occur at uneven rates across lineages and environments	Expressed and stabilized adaptation	Standing variation, selection pressure, unexpressed potential, ecological opportunity	Adaptive-landscape traversal phase	Fixation, reproductive isolation, ecological takeover	E p t e a p c t a l c

---

7.3 Embryonic development and cell differentiation

Development is one of the clearest examples of internal time differing from calendar age.

Two embryos of the same age may occupy different developmental stages.

Two cells exposed to the same signal for the same duration may differ in commitment readiness.

Developmental biology therefore already recognizes that:

Calendar age ≠ developmental state. (7.3)

The complex hypothesis adds a further claim:

> Developmental state may be organized by a conjugate relation between expressed differentiation and retained developmental potential.

A candidate cell state is:

R_cell = expressed lineage structure. (7.4)

Q_cell = remaining or competing lineage potential. (7.5)

Z_cell = R_cell + iQ_cell. (7.6)

Early in differentiation, Q may dominate.

As commitment proceeds, part of Q converts into R.

The fate gate occurs when alternative potential becomes sufficiently constrained:

Gate_fate[Z_cell] = 1. (7.7)

The event is then traced through:

- stable gene-regulatory change;
- chromatin modification;
- lineage-specific behaviour;
- loss of alternative response.

A valid complex model would need to outperform ordinary latent-state and developmental-pseudotime models.

The relevant strong test is:

ComplexPhaseModel > StandardPseudotimeModel. (7.8)

Complex notation earns priority only if conjugate phase supplies additional predictive or mechanistic value.

7.4 Homeostasis: stable R can conceal rising Q

Homeostatic systems may provide one of the strongest conceptual applications.

A physiological variable can remain within its normal range because regulatory systems are working increasingly hard to keep it there.

Examples include:

- blood glucose maintained through rising insulin output;
- blood pressure maintained through vascular and autonomic compensation;
- cardiac output maintained despite declining reserve;
- body temperature maintained through increasing metabolic cost;
- renal filtration maintained despite progressive structural loss.

The visible value R remains stable:

$R(t) \approx \text{constant}$. (7.9)

But the compensatory channel Q rises:

$Q(t) \uparrow$. (7.10)

The total state and phase therefore change:

$A(t) = \sqrt{[R^2(t) + Q^2(t)]}$. (7.11)

$\theta(t) = \text{atan2}[Q(t),R(t)]$. (7.12)

The system can move toward decompensation without obvious change in the real-axis output.

This produces a central health interpretation:

> Apparent normality is not equivalent to low internal stress.

A decompensation gate may occur when:

$\theta(t) \geq \theta_{critical}$ (7.13)

or when compensatory capacity can no longer sustain R.

This approach could distinguish:

- genuine health;
- compensated stability;
- fragile normality;
- active deterioration;
- overt failure.

However, Q must be tied to measurable regulatory effort. It cannot be inferred only after the patient fails.

7.5 Immune response as phase conversion

An immune response passes through:

- recognition;
- activation;
- expansion;
- effector action;
- regulation;
- resolution;
- memory formation.

Calendar time since exposure does not fully identify the internal immune stage.

A candidate complex state is:

R_{immune} = realized protective function. (7.14)

Q_{immune} = latent clonal capacity + antigenic pressure + unresolved inflammatory drive. (7.15)

$Z_{immune} = R_{immune} + iQ_{immune}$. (7.16)

Different outcomes may correspond to different phase trajectories:

- effective clearance;

- delayed response;
- excessive inflammation;
- immune exhaustion;
- chronic unresolved activation;
- stable memory.

The important gate is not only activation. Resolution is also a gate.

Gate_activation converts recognition into active response. (7.17)

Gate_resolution converts active response into stable memory or closure. (7.18)

A pathological system may rotate into activation but fail to pass the resolution gate.

Thus, phase alone is insufficient. The direction and gate history matter.

7.6 Ecological succession and regime change

Ecological systems often appear stable until a threshold is crossed.

A lake may remain visibly clear while nutrient loading accumulates.

A forest may retain canopy cover while regeneration capacity deteriorates.

A fish population may remain numerically stable while age structure and recruitment collapse.

A candidate ecological state is:

R_eco = visible maintained ecosystem structure. (7.19)

Q_eco = latent resilience, accumulated stress, regenerative debt, and competing regime potential. (7.20)

Z_eco = R_eco + iQ_eco. (7.21)

A regime shift occurs when a gate converts latent instability into observable structural change.

Gate_regime[Z_eco] = 1. (7.22)

The ecological ledger then includes:

- species loss;
- altered nutrient cycle;
- new trophic structure;
- changed recovery path.

After the gate, returning the visible variable R to its previous value may not restore the former world because the ledger and geometry have changed.

Therefore:

Same R_before and R_after does not imply same ecosystem. (7.23)

This is a strong example of why phase and trace must be modelled together.

7.7 Epidemics as differently paced internal worlds

Epidemics naturally separate calendar duration from propagation stage.

Different outbreaks may have:

- different transmission speeds;
- different detection delays;
- different behavioural responses;
- different intervention timing;
- different immunity structures.

Yet they may traverse comparable stages:

- latent growth;
- detectable expansion;
- healthcare pressure;
- peak;
- decline;
- endemic stabilization.

A candidate state is:

R_epi = recognized infections, recoveries, and available response capacity. (7.24)

Q_epi = latent infections, unresolved susceptibility, transmission pressure, and reporting delay. (7.25)

Z_epi = R_epi + iQ_epi. (7.26)

An outbreak declaration is one gate.

Healthcare overload is another.

Endemic transition is a third.

A phase model may allow researchers to compare fast and slow epidemics without treating calendar dates as equivalent stages.

The key test is whether:

θ_peak and θ_overload (7.27)

are more stable across outbreaks than:

t_peak and t_overload. (7.28)

7.8 The measurement challenge in biology

Biological applications face a serious risk: Q may become a vague name for hidden complexity.

To avoid this, a biological Q must satisfy:

Q is independently proxied before the gate occurs. (7.29)

Q predicts later conversion or failure. (7.30)

Q is not merely the residual of fitting R. (7.31)

Q remains interpretable across subjects or experimental conditions. (7.32)

Possible Q proxies may include:

- regulatory effort;
- reserve usage;
- latent gene-expression programmes;
- inflammatory load;
- reproductive capacity;
- habitat pressure;
- resource debt;
- unresolved pathogen burden.

A valid biological complex model must be tested against:

- ordinary multivariate state-space models;
- pseudotime methods;
- survival models;
- hidden Markov models;
- nonlinear dynamical systems;
- machine-learning latent representations.

The complex form should survive only if phase brings identifiable gain.

7.9 Interim conclusion for Groups A and B

The domains in Groups A and B share a common structure:

1. A visible or admitted state can remain stable while retained pressure changes.

2. Calendar duration does not uniquely specify internal progress.
3. Consequential transitions occur through identifiable gates.
4. Past gates alter the future state space.
5. Episodes may become more comparable after phase alignment.

The central empirical question is:

Does θ reveal a transferable internal order that t does not? (7.33)

If yes, the domain may support secondary time.

If phase also organizes gates, trace, and backreaction, the domain may support a secondary time-bearing world.

Next installment: Section 8 — Domain Group C: Infrastructure, Engineering, and Operational Networks; Section 9 — Domain Group D: Institutional, Social, and Cultural Candidates.

8. Domain Group C — Infrastructure, Engineering, and Operational Networks

8.1 Why engineering domains are especially important

Engineering and operational systems may provide the strongest early testing grounds for complex secondary-time models.

They often contain:

- dense time-series data;
- repeated episodes;
- explicit thresholds;
- measurable control inputs;
- operational gates;
- persistent logs;
- identifiable failure and recovery events.

These features make it possible to test whether phase is merely descriptive or whether it improves prediction and intervention.

The strongest calibration case is AC power engineering.

In AC systems, complex numbers are not decorative. They encode a physically meaningful relation among:

- active power;
- reactive power;

- phase difference;
- impedance;
- energy transfer;
- synchronization.

This gives an important standard:

> A new domain should not be complexified merely because two quantities can be paired. It should be complexified when phase organizes system behaviour in a way that an unrestricted pair of real variables does not express as naturally.

The engineering group therefore serves two purposes.

First, it offers promising new applications.

Second, it supplies a benchmark for judging whether those applications are genuinely complex or only rhetorically complex.

8.2 Table C — Infrastructure, Engineering, and Operational Networks

Domain	Candidate R	Candidate Q	Secondary-time interpretation	Gate or ledger event	How to test
AC power systems	Active power delivered to loads	Reactive power, quadrature support, phase displacement	Electrical phase is already a native internal coordinate	Synchronization loss, protection trip, fault clearing, blackout	Use as the calibration case: phase is independently measurable, complex transformation is physically operative, and real/reactive channels are genuinely conjugate
Power-grid stress and recovery	Delivered service, stable frequency, available generation	Reserve depletion, reactive constraint, cascading pressure, hidden fragility	Grid-stability or recovery phase	Load shedding, islanding, cascading failure, black start, service restoration	Compare blackout and restoration episodes by phase rather than elapsed time; test whether instability and recovery gates occur at stable phase regions

Domain	Candidate R	Candidate Q	Secondary-time interpretation	Gate or ledger event	How to test
Supply chains	Fulfilled orders, available inventory, realized throughput	Backlog, pipeline congestion, supplier fragility, delayed commitments, transport pressure	Fulfilment–shortage phase	Stockout, production halt, emergency sourcing, rerouting, recovery	Estimate R from realized delivery and Q from unresolved pipeline pressure; test whether shortage gates align better in θ than in days of inventory
Manufacturing flow	Completed acceptable units and realized production	Work-in-progress, queue pressure, defect burden, maintenance debt	Flow-conversion phase	Batch release, quality rejection, machine failure, line restart	Model throughput and unresolved work as conjugate channels; test whether phase predicts bottleneck and quality gates
Transport networks	Completed journeys, available network capacity, realized flow	Congestion pressure, delayed demand, route imbalance, incident stress	Congestion or network-lock phase	Gridlock, diversion, service suspension, lane closure, recovery	Compare congestion episodes of different durations after phase alignment; test whether collapse and recovery occur within repeatable phase bands
Cloud-computing platforms	Successfully served workload and verified availability	Queue depth, retries, hidden contention, memory pressure, error accumulation	Saturation–failure phase	Rate limiting, failover, outage declaration, rollback, recovery	Build R from successful service and Q from latent contention; test whether outage risk is more stable in θ than in traffic volume or elapsed load duration
Cybersecurity incidents	Verified secure operation, contained	Undetected intrusion, privilege accumulation,	Attack-progression phase	Detection, containment, credential revocation,	Re-index incidents by attack phase rather than

Domain	Candidate R	Candidate Q	Secondary-time interpretation	Gate or ledger event	How to test
	assets, confirmed access state	lateral-movement potential, unresolved exposure		breach disclosure, recovery	clock time; test whether containment gates and damage outcomes become more predictable
Materials fatigue	Remaining load-bearing performance and verified integrity	Stored microdamage, cyclic stress history, crack potential	Damage-accumulation phase	Crack initiation, propagation threshold, inspection rejection, structural failure	Compare components under different loading rates using accumulated phase; test whether failure gates align better than by calendar age
Battery ageing	Available capacity, power delivery, measured state of health	Internal resistance growth, lithium loss, latent degradation, thermal pressure	Electrochemical ageing phase	Capacity threshold, thermal event, safety shutdown, retirement	Align batteries with different charging histories by phase; test whether degradation gates and remaining life become more stable
Construction and urban development	Built and occupied infrastructure, verified project completion	Planning pressure, financing commitments, unresolved approvals, latent demand, externality burden	Development-conversion phase	Planning approval, financial close, construction start, occupancy, abandonment	Compare projects with different delays using phase and gate sequence rather than calendar duration

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8.3 AC power as the calibration case

AC power engineering demonstrates what a mature complex representation looks like.

A standard complex power quantity may be written:

$$S = P + iQ. \quad (8.1)$$

where:

P = active power. (8.2)

Q = reactive power. (8.3)

The magnitude is:

$|S| = \sqrt{(P^2 + Q^2)}$. (8.4)

The phase angle satisfies:

$\theta = \text{atan2}(Q,P)$. (8.5)

Here the imaginary component is not a vague metaphor for hidden pressure.

Reactive power has:

- independent measurement;
- operational meaning;
- physical effects;
- phase relationships;
- consequences for voltage support and system stability.

This makes AC power a powerful benchmark.

A candidate new domain should seek analogous evidence:

1. R and Q can be independently measured.
2. The two channels are structurally coupled.
3. Their phase affects system behaviour.
4. Phase provides predictive or control value.
5. The complex representation is more natural than an arbitrary real pair.

The lesson is not:

Every domain should copy electrical engineering. (8.6)

The lesson is:

Every proposed complex domain should meet an equally disciplined burden of proof. (8.7)

8.4 Power grids as multi-scale secondary worlds

A power grid contains several time structures.

There is:

- physical clock time;
- electrical phase;

- dispatch interval;
- protection-event order;
- outage and restoration ledger.

A simplified grid state may be written:

$$Z_{grid} = R_{grid} + iQ_{grid}. \quad (8.8)$$

where:

R_{grid} = realized delivered power and stable operating output. (8.9)

Q_{grid} = reactive constraint, reserve depletion, and unresolved instability pressure. (8.10)

The internal phase may move toward instability even while delivered service remains apparently normal.

Thus:

$$\frac{dR_{grid}}{dt} \approx 0 \quad (8.11)$$

can coexist with:

$$\frac{dQ_{grid}}{dt} > 0. \quad (8.12)$$

The grid appears stable on the real axis but moves toward a protection gate in complex phase.

A gate may occur when:

$$\theta_{grid} \in \Theta_{trip}. \quad (8.13)$$

or when a state-dependent threshold is crossed.

After a trip or blackout, the ledger changes the future.

The system is no longer merely at a different instantaneous state. Its:

- topology;
- available generation;
- protection configuration;
- operator action;
- restoration path;

have changed.

Therefore:

$$\text{PostGateGrid} \neq \text{PreGateGrid with lower R only.} \quad (8.14)$$

The ledger matters.

8.5 Supply chains: delivery time versus fulfilment

phase

Supply chains already distinguish among:

- ordered goods;
- goods in production;
- goods in transit;
- goods delayed;
- goods received;
- goods accepted;
- goods paid.

Calendar time alone is weak because two orders of equal age may occupy entirely different pipeline states.

A candidate supply-chain complex state is:

R_supply = realized fulfilled flow. (8.15)

Q_supply = unresolved pipeline, backlog, congestion, and supplier pressure. (8.16)

Z_supply = R_supply + iQ_supply. (8.17)

A stockout may occur not because calendar time has exceeded a fixed duration, but because the internal phase has moved into a shortage region.

Gate_stockout[Z_supply] = 1. (8.18)

Possible traces include:

- emergency procurement;
- production interruption;
- supplier substitution;
- contractual penalty;
- inventory-policy revision.

These traces alter future ordering and network structure.

A useful test would compare:

Pr(stockout | inventory days, elapsed time) (8.19)

against:

Pr(stockout | θ_supply, A_supply, network state). (8.20)

If phase improves transfer across industries or disruption types, the complex representation gains credibility.

8.6 Manufacturing flow and hidden work-in-progress

Manufacturing systems often report output while carrying hidden internal pressure.

Two factories may deliver the same number of completed units.

Yet one may have:

- low work-in-progress;
- low defect burden;
- stable maintenance;
- balanced queues.

The other may have:

- excessive work-in-progress;
- growing rework;
- fragile machinery;
- delayed inspection;
- overloaded buffers.

Therefore:

Same output does not imply same manufacturing state. (8.21)

A candidate state is:

R_mfg = accepted finished production. (8.22)

Q_mfg = unresolved queue, defect, maintenance, and rework pressure. (8.23)

Z_mfg = R_mfg + iQ_mfg. (8.24)

A phase model may reveal movement toward:

- bottleneck;
- quality failure;
- machine breakdown;
- batch rejection;
- line stoppage.

The manufacturing gate structure is especially useful because events are explicit and logged.

A phase-aware controller may choose maintenance or rerouting based on θ rather than elapsed operating hours alone.

This allows a direct intervention test:

Loss_phase-aware < Loss_calendar-based. (8.25)

8.7 Transport networks and congestion phase

Congestion is not simply the number of vehicles.

A road system can support high traffic while remaining fluid.

It can also collapse at a similar traffic count because of:

- uneven route distribution;
- lane closure;
- incident propagation;
- merging behaviour;
- signal timing;
- delayed demand.

A candidate transport state is:

R_transport = realized completed flow. (8.26)

Q_transport = delayed demand, congestion pressure, route imbalance, and unresolved blockage. (8.27)

Z_transport = R_transport + iQ_transport. (8.28)

Calendar time during rush hour is only a rough proxy.

The stronger variable may be congestion phase.

A gridlock gate occurs when:

θ_transport ≥ θ_lock. (8.29)

The resulting trace may include:

- route diversion;
- schedule breakdown;
- service suspension;
- emergency control;
- long recovery tail.

The same visible traffic count may have different future outcomes depending on phase and ledger history.

8.8 Cloud platforms: load is not saturation

Cloud systems provide another strong test environment.

A platform may serve the same request volume under two very different internal conditions.

In one case:

- queues are short;
- retries are low;
- memory is stable;
- dependencies are responsive.

In another:

- queues are building;
- retries are amplifying;
- memory pressure is rising;
- downstream services are degrading.

A candidate state is:

R_{cloud} = successfully served verified workload. (8.30)

Q_{cloud} = unresolved queue, contention, retry, and resource pressure. (8.31)

$Z_{cloud} = R_{cloud} + iQ_{cloud}$. (8.32)

The phase may move toward an outage even while the visible service level remains high.

This gives:

Observable availability ≈ stable (8.33)

while:

$\theta_{cloud} \rightarrow \theta_{fail}$. (8.34)

The event gate may be:

- rate limiting;
- failover;
- circuit-breaker activation;
- outage declaration;
- rollback.

Because cloud systems preserve detailed trace, researchers can compare:

- phase before failure;
- phase before successful intervention;

- phase before recovery.

A secondary-time model becomes valuable if it identifies equivalent internal states across outages of different duration and scale.

8.9 Cybersecurity incidents as attack-phase worlds

Cybersecurity already uses staged concepts such as:

- initial access;
- persistence;
- privilege escalation;
- lateral movement;
- exfiltration;
- detection;
- containment;
- recovery.

These stages are not evenly spaced in clock time.

An attacker may remain latent for months, then escalate in minutes.

A candidate complex state is:

R_cyber = verified secure or controlled operation. (8.35)

Q_cyber = undetected compromise, privilege accumulation, and unresolved exposure. (8.36)

Z_cyber = R_cyber + iQ_cyber. (8.37)

The internal phase may progress while R appears stable because the breach remains unobserved.

The detection gate converts hidden attack potential into institutional event:

Gate_detect[Z_cyber] = 1. (8.38)

The trace includes:

- incident classification;
- credential revocation;
- forensic record;
- legal disclosure;
- architecture change.

This is a strong example of the distinction:

Internal progression can precede public history. (8.39)

The attack may advance in phase before the organization’s ledger records the event.

8.10 Materials fatigue and battery ageing

Materials fatigue and battery ageing show why calendar age is often a poor proxy for internal state.

Two components manufactured on the same date may have very different internal damage histories.

A candidate fatigue state is:

R_fatigue = remaining verified load-bearing function. (8.40)

Q_fatigue = accumulated microdamage and crack potential. (8.41)

Z_fatigue = R_fatigue + iQ_fatigue. (8.42)

A failure gate occurs when internal damage becomes structurally propagating.

Similarly, for batteries:

R_battery = available capacity and power. (8.43)

Q_battery = latent resistance growth, material loss, and thermal pressure. (8.44)

Z_battery = R_battery + iQ_battery. (8.45)

The internal ageing phase may provide a better clock than:

- manufacturing date;
- number of calendar months;
- simple cycle count.

A genuine phase clock should align degradation trajectories across different usage histories.

The test is:

Variance[remaining life | θ] < Variance[remaining life | calendar age]. (8.46)

8.11 Construction and urban development

Construction projects are shaped by:

- planning;
- financing;
- legal approval;
- material supply;
- contractor capacity;

- public opposition;
- infrastructure dependency.

The project may remain in one calendar stage for years and then pass through several gates quickly.

A candidate state is:

R_build = verified built and usable structure. (8.47)

Q_build = unresolved planning, financing, approval, and externality pressure. (8.48)

Z_build = R_build + iQ_build. (8.49)

The project's secondary time may be ordered by:

- planning readiness;
- financial closure;
- construction commitment;
- occupancy readiness.

These stages form a gate–ledger sequence.

A project that receives planning approval is not merely older. It inhabits a new operational world because future actions become admissible.

Thus:

Gate changes the future state space. (8.50)

This is precisely the world-forming characteristic required for a stronger secondary-time interpretation.

8.12 Interim conclusion for Group C

Engineering and operational domains are especially promising because they allow the strongest empirical sequence:

Measure R and Q. (8.51)

Estimate θ. (8.52)

Predict a gate. (8.53)

Intervene before the gate. (8.54)

Observe whether the outcome changes. (8.55)

This sequence moves beyond analogy.

It tests whether phase is causally and operationally useful.

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9. Domain Group D — Institutional, Social, and Cultural Candidates

9.1 Why these domains are conceptually rich but empirically difficult

Institutional and social systems possess especially strong forms of:

- declaration;
- gate;
- public trace;
- authority;
- identity;
- memory;
- revision;
- backreaction.

A law, policy, public norm, audit opinion, or institutional decision can change the world that future observers inhabit.

These systems therefore strongly resemble secondary time-bearing worlds.

However, they are also difficult to model because:

- boundaries are contested;
- feature maps vary across observers;
- data may be politically produced;
- Q is difficult to measure independently;
- phase may be reconstructed retrospectively;
- public narratives can alter the system being measured.

The correct standard is therefore stricter.

A social complex model must declare:

$$P = (B, \Delta, h, u). \quad (9.1)$$

It must also preserve:

- observer position;
- data provenance;
- alternative frames;

- suppressed residual;
- revision conditions.

Without this discipline, the complex model can become a narrative machine.

9.2 Table D — Institutional, Social, and Cultural Candidates

Domain	Candidate R	Candidate Q	Internal phase	Gate and trace	Likely discovery method
Technology adoption	Installed use, standardized practice, realized network participation	Latent demand, infrastructure readiness, switching pressure, unresolved interoperability	Adoption-readiness or network-takeoff phase	Procurement, standard adoption, platform takeoff, institutional lock-in	Compare technologies by phase rather than years since invention; test whether adoption gates occur at stable R–Q orientations
Social movements	Public participation, visible organization, recognized demands	Latent grievance, network readiness, suppressed opposition, unexpressed support	Mobilization or legitimacy phase	Mass protest, policy concession, institutional recognition, electoral change	Reconstruct phase from public and latent network indicators; test whether large transitions align across cases
Cultural change	Publicly accepted norm, dominant narrative, ritualized practice	Counter-narratives, unresolved contradiction, generational pressure, suppressed alternatives	Norm-transition phase	Legal recognition, curriculum change, language shift, ritual adoption	Use longitudinal multi-source data; test whether normative gates occur at repeatable phase regions
Political institutions	Official policy, recognized authority, enacted decision	Coalition instability, legitimacy residual, unrepresented interests, institutional stress	Governance-stability or regime-transition phase	Election, confidence vote, constitutional change, regime transition	Compare political episodes by phase while preserving competing frames and provenance
Insurance claims	Admitted liability, recognized reserve, paid	Disputed exposure, latent loss development,	Claim-development or settlement phase	Reserve revision, settlement, judgment, claim	Estimate R from recognized liability and Q

Domain	Candidate R	Candidate Q	Internal phase	Gate and trace	Likely discovery method
	claim	litigation pressure, evidence uncertainty		closure	from incurred-but-unresolved exposure; test whether settlement gates align by phase
Accounting and audit	Recognized value, posted entries, accepted financial statements	Contingent liability, estimation uncertainty, off-ledger pressure, unresolved control weakness	Recognition or impairment phase	Provision, impairment, audit qualification, restatement	Test whether phase predicts future restatement or impairment better than account age or current ratio alone
Institutional reform	Formally adopted and operationalized rules	Legacy resistance, implementation exceptions, hidden stakeholder cost, unresolved authority conflict	Reform-internalization phase	Enactment, budget implementation, enforcement, compliance, later revision	Distinguish ceremonial adoption from operational transformation; compare reforms by phase and ledger events
Personal identity development	Publicly enacted commitments and stable self-description	Unresolved alternatives, internal conflict, unrealized capacity, memory pressure	Identity-consolidation or transition phase	Commitment, role transition, integration, rupture	Use cautious longitudinal study combining self-report, behaviour, trace, and cross-frame interpretation; remain exploratory

---

9.3 Technology adoption as phase progression

Technology adoption is often measured by calendar time since invention.

This is inadequate.

Some technologies remain marginal for decades and then scale rapidly.

Others diffuse quickly but never stabilize.

The relevant internal progression may depend on:

- infrastructure;

- cost;
- standards;
- complementary products;
- social legitimacy;
- network effects;
- regulatory acceptance.

A candidate state is:

R_{tech} = realized installed and standardized use. (9.2)

Q_{tech} = latent demand, switching pressure, and complementary readiness. (9.3)

$Z_{tech} = R_{tech} + iQ_{tech}$. (9.4)

The adoption phase may move even when current usage remains low.

A network-takeoff gate may occur when:

$\theta_{tech} \in \Theta_{takeoff}$. (9.5)

After the gate:

- suppliers invest;
- standards consolidate;
- users expect compatibility;
- institutions reorganize;
- alternatives become less viable.

The adoption ledger therefore backreacts on future adoption.

This is a genuine world-forming loop:

Adoption → infrastructure → reduced friction → further adoption. (9.6)

9.4 Social movements and mobilization phase

Social movements often appear suddenly only because the latent phase was poorly observed.

Visible participation may remain low while:

- grievance accumulates;
- network ties strengthen;
- symbols consolidate;
- opposition loses legitimacy;

- organizational capacity develops.

A candidate state is:

R_movement = visible mobilization and recognized organization. (9.7)

Q_movement = latent grievance, network readiness, and suppressed support. (9.8)

Z_movement = R_movement + iQ_movement. (9.9)

A mass-mobilization gate may then appear abrupt in calendar time but less abrupt in phase time.

However, this domain is especially vulnerable to retrospective construction.

Researchers may incorrectly define Q using information known only after the movement succeeds.

Therefore:

Q must be estimated before the gate. (9.10)

The model must also disclose:

- missing voices;
- state surveillance effects;
- media distortion;
- alternative causal explanations;
- observer position.

A social phase model without provenance is weak.

9.5 Cultural change and norm-transition time

Cultural norms often change through a long period of hidden tension followed by rapid public transition.

A norm may remain officially stable while:

- private behaviour changes;
- younger generations adopt alternatives;
- language shifts;
- contradictions accumulate;
- institutions lose enforcement capacity.

A candidate cultural state is:

R_culture = publicly admitted norm and institutionalized practice. (9.11)

Q_culture = counter-narrative pressure, unresolved contradiction, and latent alternative

practice. (9.12)

Z_culture = R_culture + iQ_culture. (9.13)

The norm-transition phase may produce gates such as:

- legal recognition;
- curriculum change;
- public ritual change;
- linguistic standardization;
- institutional abandonment.

After the gate, the cultural ledger changes future interpretation.

A practice previously treated as deviant may become ordinary.

A former norm may become residual.

Thus:

Cultural gate changes the admissibility of future meaning. (9.14)

This is a strong secondary-world characteristic.

9.6 Political institutions and legitimacy phase

Political systems possess multiple local frames:

- government;
- opposition;
- courts;
- media;
- civil service;
- citizens;
- external actors.

The same event may be described differently in each frame.

A political complex model must therefore include cross-frame discipline.

A candidate state is:

R_pol = recognized authority and enacted governance capacity. (9.15)

Q_pol = legitimacy residual, coalition instability, and unrepresented pressure. (9.16)

Z_pol = R_pol + iQ_pol. (9.17)

A regime may appear stable on the real axis while legitimacy pressure grows.

A gate may occur through:

- election;
- confidence vote;
- constitutional judgment;
- mass defection;
- emergency declaration;
- regime transition.

But political data are often produced by the same institutions whose stability is being measured.

Therefore, a mature protocol must include:

Objectivity_P = CrossFrameInvariance_P + AccessibleTrace_P + ResidualAudit_P. (9.18)

This follows the broader world-formation principle that objectivity is not the absence of observers, but the preservation of governed relations across admissible frames.

9.7 Insurance claims as a particularly measurable institutional domain

Insurance may offer one of the strongest Group D test cases because it contains explicit:

- claims;
- reserves;
- evidence;
- settlement;
- litigation;
- closure records.

A candidate state is:

R_claim = admitted and recognized liability. (9.19)

Q_claim = disputed exposure, latent loss development, and unresolved litigation pressure. (9.20)

Z_claim = R_claim + iQ_claim. (9.21)

Claim age is often a weak predictor because claims develop differently.

The relevant internal order may be:

- evidence maturity;

- liability recognition;
- reserve adequacy;
- litigation pressure;
- settlement readiness.

A settlement gate occurs when:

Gate_settle[Z_claim] = 1. (9.22)

The ledger then affects:

- reserves;
- premium models;
- legal strategy;
- future underwriting;
- regulatory reporting.

Insurance therefore has clear phase, gate, trace, and backreaction structures.

9.8 Accounting and audit: recognition as a world-forming gate

Accounting is not merely passive description.

Accounting recognition determines what becomes:

- asset;
- liability;
- income;
- loss;
- provision;
- impairment;
- capital.

A candidate state is:

R_acct = recognized and posted financial structure. (9.23)

Q_acct = contingent obligation, estimation uncertainty, control weakness, and off-ledger pressure. (9.24)

Z_acct = R_acct + iQ_acct. (9.25)

An impairment or provision gate transforms unresolved pressure into recognized trace.

Gate_recognition[Z_acct] = 1. (9.26)

After recognition:

- ratios change;
- contracts may trigger;
- management action changes;
- market perception shifts;
- regulatory consequences follow.

Therefore:

Accounting gate changes the primary financial world. (9.27)

This is a direct example of secondary-world backreaction.

The accounting model does not merely observe the company.

Once gated into official trace, it alters the company's future possibilities.

9.9 Institutional reform: ceremonial versus operational phase

Institutional reform often fails because enactment is confused with completion.

A new rule may be announced, but:

- budgets remain unchanged;
- incentives remain unchanged;
- old workflows persist;
- exceptions multiply;
- enforcement is absent.

A candidate state is:

R_reform = formally adopted and operationalized structure. (9.28)

Q_reform = implementation resistance, legacy constraint, and unresolved stakeholder cost. (9.29)

Z_reform = R_reform + iQ_reform. (9.30)

The reform phase distinguishes:

- symbolic declaration;
- administrative preparation;
- budget commitment;

- operational adoption;
- enforcement;
- internalization;
- revision.

A mature reform gate requires trace:

Gate_reform = decision + authority + resource + implementation record + residual register. (9.31)

This prevents the institution from declaring success merely because a policy document exists.

9.10 Personal identity development as an exploratory boundary case

Personal identity may also possess secondary-time characteristics.

Calendar age does not determine:

- commitment;
- self-integration;
- role transition;
- maturity;
- unresolved conflict.

A candidate state might be:

R_self = publicly enacted commitments and stable self-description. (9.32)

Q_self = unresolved alternatives, unrealized capacity, memory pressure, and internal conflict. (9.33)

Z_self = R_self + iQ_self. (9.34)

Gates may include:

- commitment;
- separation;
- role transition;
- public declaration;
- integration;
- rupture.

However, this domain requires exceptional caution.

The observer and the observed system overlap.

Measurement may change the identity process itself.

The feature map may impose a theory of selfhood rather than discover one.

Therefore, any complex model of personal identity should remain exploratory and pluralistic.

It must not reduce a person to a single phase angle.

9.11 The central institutional danger: protocol power

In institutional systems, the authority to define R and Q is itself a form of power.

An organization may define:

R = official success. (9.35)

and classify all criticism as:

Q = unresolved noise. (9.36)

A government may treat dissent as residual rather than evidence.

A company may classify hidden cost as externality.

An accounting regime may postpone recognition.

An AI platform may define acceptable trace while suppressing uncertainty.

Therefore:

Power_P = capacity to control boundary, feature map, gate, trace, residual, and revision. (9.37)

This principle is central to observer-compatible world formation. A protocol does not merely describe a world; it controls what becomes visible, admissible, recorded, unresolved, and revisable.

A mature complex model must therefore disclose:

- who defines R;
- who estimates Q;
- who controls the gate;
- who owns the ledger;
- who can challenge the phase interpretation;
- what residual remains suppressed.

9.12 Evidence ranking within Group D

The Group D domains should not be treated equally.

Stronger empirical candidates

- insurance claims;
- accounting and audit;
- technology adoption;
- institutional reform.

These contain comparatively explicit records and gates.

Intermediate candidates

- political institutions;
- social movements;
- cultural change.

These contain rich event and trace structure but high observer dependence.

Exploratory candidates

- personal identity development;
- civilization-scale cultural phase;
- collective consciousness.

These may be conceptually valuable but require much stronger safeguards against imposed narrative.

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10. Cross-Domain Synthesis of Groups A–D

The four domain groups differ in substance, mechanism, data quality, and evidential maturity.

Yet they share a possible structure.

A visible state R is insufficient.

A retained conjugate channel Q influences future evolution.

The state rotates through a phase θ.

Calendar duration varies across episodes.

Consequential events occur through gates.

Gates write trace.

Trace changes future admissibility.

This yields the shared architecture:

Parent field

→ declared protocol
→ admitted state R

- retained conjugate Q
  → complex phase θ
  → accumulated internal time τ_i
  → event gate
  → persistent trace
  → parent-world backreaction. (10.1)

The domain tables are therefore not claims of completed theory.

They are search maps.

Their purpose is to identify where researchers might ask:

1. Is there a meaningful R?
2. Is there an independently measurable Q?
3. Do R and Q form a conjugate relation?
4. Does phase align differently paced episodes?
5. Do gates cluster by phase?
6. Does trace change the future?
7. Does the complex model outperform real alternatives?

Only domains that pass this sequence should be described as secondary time-bearing worlds.

Next installment: Part IV — How Secondary-Time Conversion Could Be Discovered; including the full cross-domain research protocol, model comparison tests, phase-alignment criteria, gate-locking tests, and reduction rules.

Part IV — How Secondary-Time Conversion Could Be Discovered

11. The Cross-Domain Discovery Protocol

11.1 Begin from unequal duration, not from complex notation

The research process should not begin by writing:

$Z = R + iQ$. (11.1)

It should begin by identifying a domain in which calendar duration appears to be a poor description of internal progress.

The strongest initial pattern is:

- comparable initial conditions;

- comparable consequential endpoints;
- radically different elapsed durations;
- apparently similar intermediate stages.

Examples include:

- two credit crises reaching default at different speeds;
- two patients reaching the same recovery stage over different periods;
- two legal cases reaching judgment through different schedules;
- two software projects reaching release readiness after different numbers of iterations;
- two epidemics reaching capacity overload at different transmission speeds.

Let episode j have external duration T_j.

A candidate secondary-time domain should contain episodes for which:

$T_1 \neq T_2$ (11.2)

while some internal progression may remain comparable:

$\tau_{i,1,\text{end}} \approx \tau_{i,2,\text{end}}$. (11.3)

The hypothesis is that unequal calendar durations may correspond to approximately equal internal phase distances.

11.2 Stage 1 — Declare the protocol

Every proposed conversion must begin with a declared protocol:

$P = (B, \Delta, h, u)$. (11.4)

where:

- B is the system boundary;
- Δ is the observation or aggregation rule;
- h is the time or state window;
- u is the admissible intervention family.

The protocol should also state:

- baseline q;
- feature map φ;
- gate rule G;
- trace rule T;
- residual rule R_res;

- revision condition U_{adm}.

The declared world is:

$$World_P = (X, q, \varphi, B, \Delta, h, u, G, T, R_{res}, U_{adm}). \quad (11.5)$$

Without this declaration, R and Q may drift in meaning between episodes.

For example, “bank liquidity” may refer to:

- reported liquid assets;
- immediately transferable cash;
- collateral-eligible securities;
- central-bank-accessible assets;
- stress-adjusted liquidity.

The phase derived from one definition cannot be compared honestly with the phase derived from another unless a transport map is supplied.

Therefore:

No stable phase without stable declaration. (11.6)

11.3 Stage 2 — Define the real-side coordinate R

R should represent what the system currently admits as realized, observable, verified, or committed.

A valid R should satisfy:

1. It has an operational definition.
2. It can be measured without knowledge of the later outcome.
3. It remains comparable across repeated episodes.
4. It corresponds to a recognized state under protocol P.
5. Its measurement uncertainty is disclosed.

Examples include:

- settled financial value;
- verified software functionality;
- admitted legal fact;
- transferable educational competence;
- clinically observed physiological function;
- delivered power;

- fulfilled supply-chain output;
- officially implemented policy.

R should not simply be “what looks real.”

It should be what has passed the domain’s relevant observation or admission rule.

Formally:

R_P(t) = Admit_P[Project_P(X_t)]. (11.7)

11.4 Stage 3 — Identify the conjugate coordinate Q

The candidate Q must be more than an unmeasured remainder.

A defensible Q should meet five conditions.

Condition 1 — Distinctness

Q must not be a disguised rescaling of R.

Corr(R,Q) may be high, but Q must carry distinct information. (11.8)

Condition 2 — Forward relevance

Q should predict later movement in R, gate probability, or residual formation.

Pr[R(t + Δt) | R(t),Q(t)] should differ from Pr[R(t + Δt) | R(t)]. (11.9)

Condition 3 — Conversion or compensation

Part of Q should convert into, support, oppose, defer, or regulate R.

Condition 4 — Independent estimation

Q must be estimated from data available before the event gate.

Condition 5 — Cross-episode meaning

The same Q definition should remain meaningful across different episodes.

Candidate Q measures may include:

- implied risk;
- compensatory physiological effort;
- unresolved dependency pressure;
- latent demand;
- queue depth;
- hidden damage;

- disputed legal exposure;
- unresolved scientific anomaly;
- unverified AI hypotheses.

The key distinction is:

Q_P = declared conjugate structure. (11.10)

$Residual_P$ = what remains unexplained after the R–Q model. (11.11)

Q must not absorb the entire residual.

11.5 Stage 4 — Choose the complex normalization

The raw variables R and Q may use different units or scales.

Before writing:

$Z = R + iQ$, (11.12)

the researcher must specify a normalization or metric.

A simple normalized form is:

$r = (R - \mu_R)/\sigma_R$. (11.13)

$q = (Q - \mu_Q)/\sigma_Q$. (11.14)

Then:

$Z = r + iq$. (11.15)

But standardization alone may be insufficient. Domain structure may require a weighted metric:

$A^2 = g_{RR} R^2 + 2g_{RQ} RQ + g_{QQ} Q^2$. (11.16)

If the axes are not orthogonal under the natural domain geometry, the researcher must explain why the chosen transformation produces a legitimate complex structure.

The simplest complex form assumes:

$g_{RQ} = 0$. (11.17)

$g_{RR} = g_{QQ}$ after normalization. (11.18)

This assumption must be tested rather than silently imposed.

A misleading normalization can manufacture an apparently stable phase.

Therefore:

Phase stability must survive reasonable scaling choices. (11.19)

11.6 Stage 5 — Test for phase-generating dynamics

Once R and Q are defined, estimate:

$$\theta(t) = \operatorname{atan2}[Q(t),R(t)]. \quad (11.20)$$

The first test is whether the phase has a coherent trajectory.

A candidate phase should exhibit:

- interpretable movement;
- reasonable continuity;
- identifiable branches;
- limited sensitivity to measurement noise;
- repeatable relation to domain events.

The strongest idealized dynamics are:

$$dR/d\theta \approx -Q. \quad (11.21)$$

$$dQ/d\theta \approx R. \quad (11.22)$$

But real domains may include damping, forcing, growth, and nonlinear coupling.

A more general local model is:

$$dZ/dt = [g_A(t) + i\omega(t)]Z(t) + F(Z,t) + \varepsilon(t). \quad (11.23)$$

where:

- g_A controls amplitude growth or decay;
- ω controls angular progression;
- F represents nonlinear domain coupling;
- ε is unresolved residual.

The important comparison is whether phase reparameterization simplifies the system:

$$\text{Complexity}[dZ/d\theta] < \text{Complexity}[dZ/dt]. \quad (11.24)$$

Complexity may be measured through:

- number of fitted parameters;
- coefficient instability;
- residual variance;
- model description length;
- out-of-sample error;
- cross-episode transfer failure.

11.7 Stage 6 — Construct a monotonic internal coordinate

If θ progresses monotonically within the relevant regime:

$\tau_i(t) = \text{unwrap}[\theta(t)]$. (11.25)

If phase reverses, use cumulative distance:

$\tau_i(t) = \int_0^t |\dot{\theta}(s)|\, ds$. (11.26)

but retain orientation:

$o(t) = \text{sgn}[\dot{\theta}(t)]$. (11.27)

A more complete state is:

$T_i(t) = [\tau_i(t), o(t), b(t)]$. (11.28)

where b is the branch index.

The researcher should not force a single global clock when the domain requires local charts.

A valid result may be:

- locally monotonic secondary time;
- regime-specific phase clocks;
- multiple branches;
- recurrent loops with distinct ledger histories.

This is not a defect. It may reveal the genuine topology of the secondary world.

11.8 Stage 7 — Align repeated episodes by phase

Let $x_j(t)$ denote episode j.

Ordinary analysis compares:

$x_1(t), x_2(t), \ldots, x_n(t)$. (11.29)

Secondary-time analysis compares:

$x_1[\tau_i], x_2[\tau_i], \ldots, x_n[\tau_i]$. (11.30)

The phase hypothesis gains support when:

$\text{Var}_j[x_j(\tau_i)] < \text{Var}_j[x_j(t)]$. (11.31)

This means differently paced episodes become more structurally similar after internal-time alignment.

Possible measures include:

- dynamic time-warping distance;
- phase-conditioned variance;
- trajectory clustering;
- cross-episode prediction;
- event-hazard consistency;
- transfer of fitted parameters.

A successful phase coordinate should compress episode diversity without erasing meaningful residual.

11.9 Stage 8 — Test phase-sensitive gates

A secondary clock becomes operationally important when consequential events occur at reproducible phase positions.

Let G_k be a gate event.

The central test is:

$$\Pr(G_k = 1 \mid \theta, A, R, Q) > \Pr(G_k = 1 \mid t, R, Q). \quad (11.32)$$

The gate may occur within a band:

$$\theta \in \Theta_G = [\theta_-, \theta_+]. \quad (11.33)$$

Examples include:

- default;
- decompensation;
- software release;
- judgment;
- scientific replication;
- technology takeoff;
- outage;
- cell-fate commitment.

The researcher should estimate:

- mean gate phase;
- circular variance;
- phase-window width;
- branch dependence;

- false-positive rate;
- cross-episode stability.

A narrow phase band is not always required. Some domains may have state-dependent gate surfaces:

G[Z,A,P] = 1. (11.34)

The important requirement is that phase adds stable information beyond elapsed duration.

11.10 Stage 9 — Separate phase progression from ledger events

A system may pass through many internal phase states without producing a committed event.

Therefore:

θ progression ≠ ledger advancement. (11.35)

Let:

e_k = Gate_P[Z(τ_i)]. (11.36)

The ledger then updates:

L_{k+1} = U_L(L_k,e_k,r_k). (11.37)

where r_k is the residual attached to the event.

A mature ledger record should preserve:

- event identity;
- protocol version;
- gate criterion;
- evidence;
- authority;
- phase state;
- residual;
- revision path.

The ledger is not merely an event counter.

It is past consequence carried into future admissibility.

This is consistent with Observer-Compatible World Formation, where trace becomes historical only when it affects future projection, gate, or action.

11.11 Stage 10 — Test backreaction

The strongest secondary-world case occurs when gated outputs alter the parent system.

Let the secondary-world state be Z_S.

Let the parent-world state be X_P.

A gated event produces an intervention:

$u_k = Policy[Z_S,L_k]$. (11.38)

The parent world then changes:

$X_{P,k+1} = F_P(X_{P,k},u_k,\xi_k)$. (11.39)

The changed parent world generates a new secondary state:

$Z_{S,k+1} = Project_P[X_{P,k+1}]$. (11.40)

This creates the reflexive loop:

Parent world → secondary phase → gate → ledger → intervention → changed parent world. (11.41)

Examples include:

- risk valuation changing hedge demand;
- accounting recognition changing financing capacity;
- diagnosis changing treatment;
- legal judgment changing future admissibility;
- AI recommendation changing user behaviour;
- scientific publication changing funding and experiment design;
- technology adoption changing infrastructure.

Backreaction distinguishes a passive internal coordinate from an operational secondary world.

---

12. Does Complex Structure Add More Than Two Real Variables?

12.1 The weak comparison

A weak study compares:

Model(R) (12.1)

with:

Model(R,Q). (12.2)

If the second model performs better, the result proves only that Q contains useful information.

It does not prove that the complex structure is necessary.

12.2 The strong comparison

The necessary comparison is:

FlexibleRealModel(R,Q) (12.3)

against:

ComplexPhaseModel(R + iQ). (12.4)

The flexible real model may include:

- nonlinear interactions;
- arbitrary two-dimensional vector fields;
- hidden states;
- time-varying coefficients;
- neural dynamics;
- state-space estimation.

The complex model earns priority only if its constrained structure produces a gain.

Possible gains include:

- lower parameter count;
- greater interpretability;
- better phase alignment;
- more stable transfer;
- stronger gate prediction;
- improved intervention;
- better cross-frame invariance;
- clearer reduction rules.

The criterion is:

Benefit_complex > Cost_complex. (12.5)

12.3 Model comparison ladder

A mature study should compare at least five models.

Model 0 — Calendar-only baseline

$Y(t) = f(t)$. (12.6)

Model 1 — Real observable model

$Y(t) = f[R(t)]$. (12.7)

Model 2 — Flexible two-real-variable model

$Y(t) = f[R(t),Q(t)]$. (12.8)

Model 3 — Complex phase model

$Y(\tau_i) = f[A(\tau_i),\theta(\tau_i)]$. (12.9)

Model 4 — Phase–gate–ledger model

$Y_{k+1} = f[A_k,\theta_k,L_k,r_k,u_k]$. (12.10)

The research question is not whether Model 4 always wins.

It is whether each additional layer earns its complexity.

12.4 Phase compression

Define calendar-aligned dispersion:

$D_t = \Sigma_j \int ||x_j(t) - \bar{x}(t)||^2 \, dt$. (12.11)

Define phase-aligned dispersion:

$D_\theta = \Sigma_j \int ||x_j(\theta) - \bar{x}(\theta)||^2 \, d\theta$. (12.12)

Phase compression occurs when:

$D_\theta < D_t$. (12.13)

A strong result should hold:

- out of sample;
- across regimes;
- under reasonable normalization;
- across admissible observers.

If phase alignment works only after case-specific tuning, it may be retrospective overfitting.

12.5 Gate concentration

For event phases $\theta_1, \theta_2, \ldots, \theta_n$, define the circular concentration:

$C_G = |n^{-1} \Sigma_j \exp(i\theta_j)|$. (12.14)

Then:

$0 \le C_G \le 1$. (12.15)

A value near 1 indicates strong phase concentration.

A value near 0 indicates widely dispersed event phases.

But high concentration alone is insufficient. The result must outperform:

- calendar-time hazard;
- direct R and Q thresholds;
- conventional domain indicators;
- flexible nonlinear classifiers.

12.6 Dynamic simplification

Suppose the calendar-time model requires time-varying coefficients:

$dZ/dt = a(t)Z + b(t)Z^* + \varepsilon(t)$. (12.16)

while the phase-indexed model is approximately:

$dZ/d\theta = iZ + \eta(\theta)$. (12.17)

If η is small and stable, phase has simplified the dynamics.

A possible simplification score is:

$S_{dyn} = MDL_t - MDL_\theta$. (12.18)

where MDL is model description length.

Then:

$S_{dyn} > 0$ (12.19)

supports the phase representation.

12.7 Predictive gain

Let $Loss_t$ denote out-of-sample prediction loss using calendar-time models.

Let $Loss_\theta$ denote loss using phase-time models.

Define:

$G_{pred} = Loss_t - Loss_\theta$. (12.20)

A positive value supports phase usefulness.

Prediction targets may include:

- gate probability;
- remaining time to event;
- failure severity;
- recovery outcome;
- intervention response;
- residual accumulation.

The gain should be tested across multiple episodes rather than only within one trajectory.

12.8 Intervention gain

A particularly strong test compares actions chosen by calendar time with actions chosen by phase.

Let policy π_t intervene according to t-based thresholds.

Let policy π_θ intervene according to phase.

Define:

$$G_{int} = E[Utility \mid \pi_\theta] - E[Utility \mid \pi_t]. \quad (12.21)$$

Examples include:

- earlier bank-liquidity intervention;
- phase-aware maintenance;
- phase-aware rehabilitation progression;
- AI verification timing;
- supply-chain rerouting;
- epidemic-control adjustment.

A model that improves description but cannot improve action may still be scientifically useful, but its world-forming significance is weaker.

12.9 Cross-frame robustness

Different observers may estimate R and Q differently.

Let frame α produce:

$$Z^{(\alpha)} = R^{(\alpha)} + iQ^{(\alpha)}. \quad (12.22)$$

Let frame β produce:

$Z^{(\beta)} = R^{(\beta)} + iQ^{(\beta)}$. (12.23)

A mature framework requires a transport map:

$T_{\alpha\to\beta}: Z^{(\alpha)} \to Z^{(\beta)}$. (12.24)

The coordinates may differ, but event relations should remain coherent.

Possible invariants include:

- gate order;
- phase ordering;
- event identity;
- accumulated internal depth;
- trace consistency;
- intervention consequence.

Objectivity is therefore not identical phase values in every frame.

It is preserved governed relation under admissible transformation.

This follows the protocol-bound principle:

Objectivity_P = cross-frame invariance + accessible trace + residual audit. (12.25)

---

13. Evidence Levels and Claim Discipline

13.1 Descriptive success

A descriptive model shows that R and Q can be represented intuitively through magnitude and phase.

This may aid visualization and communication.

But descriptive success does not establish internal time.

Appropriate claim:

> The domain admits a useful complex representation under protocol P.

13.2 Dynamical success

A dynamically successful model shows that phase:

- simplifies equations;
- improves prediction;
- aligns episodes;

- identifies meaningful regimes.

Appropriate claim:

> Complex phase provides a useful internal progression coordinate.

13.3 Secondary-time success

A secondary-time model shows that differently paced episodes become comparable under τ_i.

Appropriate claim:

> The domain supports a locally useful secondary phase time.

This statement should disclose:

- applicable regime;
- branch structure;
- monotonicity conditions;
- normalization;
- residual.

13.4 Gate-bearing success

A stronger result shows that consequential events occur within stable phase regions.

Appropriate claim:

> Domain events are phase-sensitive under the declared protocol.

13.5 World-forming success

The strongest result adds:

- persistent trace;
- future constraint;
- observer update;
- parent-world backreaction;
- admissible revision.

Appropriate claim:

> The domain supports a ledgered secondary time-bearing world interpretation.

This is the highest claim level and should require the strongest evidence.

13.6 Recommended evidence ladder

Level	Required evidence	Recommended language
E1 — Pairing	R and Q are useful distinct variables	"Two-channel representation"
E2 — Complex completion	Magnitude and phase are operationally meaningful	"Complex-state model"
E3 — Phase dynamics	θ simplifies or predicts evolution	"Phase-bearing dynamics"
E4 — Secondary order	Unequal-duration episodes align in τ_i	"Local secondary time"
E5 — Phase gate	Consequential events concentrate by phase	"Phase-sensitive event formation"
E6 — Ledgered world	Gates write trace and alter future dynamics	"Secondary time-bearing world"

The article's domain tables contain candidates across these levels. They should not be read as claims that every domain has already reached E6.

Part V — Interpretation, Limits, and Research Position

14. Relationship to *When Valuation Becomes a World, Part II*

14.1 Finance as the originating model case

When Valuation Becomes a World, Part II provides the detailed architecture from which the broader proposal is abstracted.

Its central move is not merely to write financial value as a complex number.

It constructs a layered financial world containing:

- local valuation state;
- retained risk or valuation pressure;
- phase progression;
- derivative composite states;
- internal observers;
- local frames;
- global geometry;
- gates;
- ledgers;
- backreaction.

The financial case demonstrates how a secondary model world can become operationally real.

A valuation affects:

- execution;
- margin;
- hedging;
- collateral;
- capital;
- accounting;
- regulation;
- settlement.

Once these gates are crossed, the resulting trace alters future market behaviour.

The financial model therefore supplies the full chain:

Primary financial conditions
→ complex valuation state
→ phase progression
→ institutional gate
→ ledgered event
→ changed future financial environment. (14.1)

The present article generalizes that chain.

14.2 What is being transferred

The article does not transfer financial variables literally into biology, law, AI, or ecology.

It transfers a research grammar:

1. identify admitted structure;
2. identify a conjugate retained channel;
3. construct a phase-bearing state;
4. compare internal phase with external duration;
5. identify event gates;
6. trace committed consequences;
7. test backreaction;
8. preserve residual;
9. reduce the model when added structure provides no gain.

The cross-domain object is therefore not “finance everywhere.”

It is:

Phase–gate–ledger world formation under bounded protocols. (14.2)

14.3 Finance as one member of a broader class

If the programme succeeds, finance would no longer be an isolated metaphorical application of complex numbers.

It would become one member of a wider class:

- valuation worlds;
- developmental worlds;
- procedural worlds;
- reasoning worlds;
- operational worlds;
- institutional worlds.

Each contains its own:

- projection;
- phase;
- gates;
- trace;
- internal ordering;
- residual;
- parent-world interaction.

This is the broader significance of Part II.

Its architecture may serve as a prototype for studying other mature domains whose internal progress cannot be reduced to calendar duration.

---

15. Relationship to Observer-Compatible World Formation

15.1 The outer grammar

Observer-Compatible World Formation proposes that a stable world requires:

- boundary;
- identity;

- mediation;
- binding;
- gate;
- trace;
- residual;
- invariance;
- admissible revision.

Its core cycle is:

$\Sigma_0 \rightarrow \text{Declare_P} \rightarrow \text{Project_P} \rightarrow \text{Gate_P} \rightarrow \text{Trace_P} + \text{Residual_P} \rightarrow \text{InvarianceTest_P} \rightarrow \text{Revise_P} \rightarrow \Sigma'$. (15.1)

This architecture explains how a field becomes usable, historical, comparable, and revisable for bounded observers.

15.2 Complex phase as an inner clock

The present article adds a narrower proposal.

Between projection and gate, some domains may support a complex state:

$\text{Project_P}(\Sigma_P) \rightarrow Z_P = R_P + iQ_P$. (15.2)

Its phase may generate internal progression:

$Z_P \rightarrow \theta_P \rightarrow \tau_i,P$. (15.3)

The completed cycle becomes:

Σ_0
$\rightarrow$ Declare_P
$\rightarrow$ Project_P
$\rightarrow$ R_P + iQ_P
$\rightarrow \theta_P$
$\rightarrow \tau_i,P$
$\rightarrow$ Gate_P
$\rightarrow$ Trace_P + Residual_P
$\rightarrow$ InvarianceTest_P
$\rightarrow$ Revise_P
$\rightarrow \Sigma'$. (15.4)

Thus:

OCWF supplies the world-forming runtime. (15.5)

Complex phase may supply the internal clock. (15.6)

Not every observer-compatible world requires complex secondary time.

But some may obtain their internal order through phase.

15.3 Relationship to protocol-bound isomorphism

From Interfaces to Isomorphisms argues that cross-domain transfer becomes disciplined only when role correspondences are declared under protocol rather than asserted as substance identities.

This principle is essential here.

The valid form is not:

Biological Q = financial Q. (15.7)

The valid form is:

Biological Q and financial Q may perform corresponding conjugate-retention roles under their respective declared protocols. (15.8)

Likewise:

Legal gate ≠ market settlement gate. (15.9)

But both may convert unresolved possibility into trace-bearing institutional consequence.

The proposed cross-domain extension is therefore an isomorphism programme, not a universal substance theory.

---

16. Failure Conditions and Reduction Rules

16.1 No meaningful magnitude

If A has no independent operational meaning, the polar form may be arbitrary.

Reject or revise the complex interpretation when:

$A^2 = R^2 + Q^2$ (16.1)

exists only because the researcher imposed a convenient normalization.

16.2 Q is an error bucket

Reject the model when Q is defined as:

Q = everything not explained by R. (16.2)

A valid Q requires:

- prior definition;
- independent proxy;
- forward relevance;
- stable domain meaning.

Unexplained remainder should remain residual.

16.3 No phase simplification

If:

dZ/dθ (16.3)

is not simpler, more stable, or more predictive than:

dZ/dt, (16.4)

phase should not be promoted to an internal clock.

16.4 No episode compression

If differently paced episodes remain equally dissimilar after phase alignment:

$D_\theta \geq D_t$, (16.5)

the candidate secondary time has failed an important test.

16.5 No gate concentration

If event gates are not more stable in phase than in calendar time or direct state variables, phase may be descriptive but not event-bearing.

16.6 Phase depends strongly on arbitrary scaling

If small changes in normalization radically alter phase order, the complex state lacks robustness.

16.7 No gain over flexible real models

If:

Performance_complex ≤ Performance_real-pair, (16.6)

retain the two-dimensional real model.

The ordered pair may be fully sufficient.

16.8 No gain from Q

If Q adds no predictive, diagnostic, or intervention value:

Return from $Z = R + iQ$ to R. (16.7)

16.9 No trace or backreaction

A domain may contain a useful phase clock without forming a complete secondary world.

In that case, use the weaker description:

> phase-indexed internal dynamics

rather than:

> secondary time-bearing world.

16.10 Claim reduction ladder

A mature theory should be able to reduce itself.

Full phase–gate–ledger world
→ remove backreaction
→ phase-bearing event model
→ remove ledger
→ secondary phase-time model
→ remove gate
→ complex dynamical model
→ remove phase privilege
→ two-real-variable model
→ remove Q
→ one-variable model. (16.8)

The correct model is the least complex level that retains explanatory and predictive gain.

---

17. The Calculus Analogy

17.1 Why the analogy is attractive

Calculus supplied a portable language for:

- local change;
- rate;
- accumulation;
- curvature;
- optimization.

It could travel across physics, engineering, economics, biology, and control because those domains all contain change and accumulation.

The present proposal asks whether protocol-bound complexification could supply a portable language for:

- conjugate completion;

- phase progression;
- uneven internal speed;
- gate-sensitive conversion;
- ledgered consequence.

In compressed form:

Calculus asks how much a state changes. (17.1)

Complex secondary-time dynamics asks how a state rotates between admitted and retained structure, and when that rotation becomes history. (17.2)

17.2 Why the analogy remains preliminary

Calculus possesses:

- precise universal operators;
- mature theorems;
- standardized notation;
- centuries of empirical use;
- broad independent validation.

The proposed framework currently possesses:

- a conceptual architecture;
- candidate role mappings;
- testable comparison criteria;
- multiple possible applications.

It does not yet possess a universally established measurement or theorem structure.

Therefore, the proper claim is not:

> A new calculus has already been discovered.

The proper claim is:

> A possible cross-domain research programme has been identified in which complex phase may play a role analogous to a portable internal-ordering operator.

17.3 What would make the analogy stronger

The calculus analogy would become substantially stronger if independent studies demonstrated that:

1. the same formal complexification protocol works across multiple domains;

2. phase alignment improves episode comparison;
3. gate phases remain stable;
4. complex models outperform flexible real alternatives;
5. phase-aware intervention improves outcomes;
6. a shared theorem or invariance structure emerges.

At that stage, one could begin to speak of:

- Protocol-Bound Complex Dynamics;
- Complex Secondary-Time Calculus;
- Phase–Gate–Ledger Mathematics;
- Complex World-Formation Dynamics.

For now, these are programme names rather than completed disciplines.

---

18. Conclusion — When Phase Becomes a Clock

At one static instant:

R + iQ (18.1)

contains the same numerical information as:

(R,Q). (18.2)

The deeper reason to prefer complex representation is therefore not storage.

It is phase.

A complex state carries a canonical generator of rotation. Under suitable conditions, irregular external evolution can be reparameterized as a simpler internal progression:

dZ/dt = [g_A + iω]Z + ε. (18.3)

dZ/dθ ≈ iZ. (18.4)

The parent world may observe:

- acceleration;
- delay;
- plateau;
- reversal;
- crisis.

The secondary world may interpret the same trajectory as unequal calendar speeds through a comparable phase path.

But phase alone is not yet time-bearing world formation.

The strongest candidate requires:

Uneven parent duration

- independently measurable conjugate state
- robust phase order
- phase-sensitive event gates
- persistent trace
- future constraint
- parent-world backreaction. (18.5)

This article has mapped that possibility across four domain groups:

- finance, law, AI, projects, science, education, and medicine;
- biology and ecology;
- engineering, infrastructure, and operational networks;
- institutions, society, culture, and identity.

These examples are not assertions that every domain already possesses imaginary time.

They are search maps.

The research question is:

> Can differently paced episodes be represented by a conjugate complex state whose phase provides a more stable internal order than calendar duration?

The stronger question is:

> Do consequential events occur at reproducible phase positions, enter persistent trace, and alter the future world?

Where the answer is yes, R + iQ may be more than an elegant representation.

It may reveal how a projected real state acquires:

- orientation;
- internal depth;
- event structure;
- historical order;
- world-forming consequence.

The final proposal can therefore be stated carefully:

> The imaginary coordinate should not be assigned one universal substance meaning. Its possible cross-domain importance is structural. It may supply the conjugate

completion through which an admitted real projection becomes phase-bearing. Where that phase generates a more stable internal order than calendar time, predicts consequential gates, and enters persistent trace, complexification may reveal a genuine secondary time-bearing world.

In its most compact form:

Complex completion → phase → internal time → gate → trace → world. (18.6)

Appendix A — Cross-Domain Master Index

This appendix consolidates the four discovery groups into a compact research map. The detailed domain tables remain in Sections 6–9.

The purpose of the index is not to rank the importance of the domains. It ranks their likely usefulness for uncovering a complex secondary-time structure.

A.1 Domain-group comparison

Group	Domain character	Main strength	Main difficulty	Present research priority
A — Highest-priority candidates	Finance, law, AI, projects, science, education, medicine	Clear gates, records, repeated episodes, unequal durations	Defining a non-arbitrary conjugate Q	Very high
B — Biological and ecological candidates	Development, compensation, immunity, ecosystems, epidemics	Strong internal stages and hidden regulatory effort	Q may be difficult to measure independently	High
C — Engineering and operational networks	Power, infrastructure, software, supply chains, fatigue	Dense data, controlled interventions, explicit failures	Phase may remain system-specific rather than universal	Very high
D — Institutional and cultural candidates	Adoption, reform, politics, culture, identity	Strong gate–trace–backreaction structure	Observer dependence and protocol power	Exploratory to high, depending on domain

A.2 Candidate-domain inventory

Group A — Highest-priority domains

1. Derivatives and credit markets
2. Bank balance-sheet stress
3. Project and programme delivery
4. Organizational transformation
5. Legal proceedings

6. AI reasoning and agent runtimes
7. Software delivery and DevOps
8. Scientific research programmes
9. Education and skill acquisition
10. Clinical disease progression
11. Recovery and rehabilitation
12. Neural decision formation

Group B — Biological and ecological domains

1. Embryonic development
2. Cell differentiation
3. Immune response
4. Homeostasis and physiological compensation
5. Metabolic adaptation
6. Ecological succession
7. Population collapse and recovery
8. Epidemics
9. Evolutionary transitions

Group C — Engineering and operational domains

1. AC power systems
2. Power-grid stress and recovery
3. Supply chains
4. Manufacturing flow
5. Transport networks
6. Cloud-computing platforms
7. Cybersecurity incidents
8. Materials fatigue
9. Battery ageing
10. Construction and urban development

Group D — Institutional and cultural domains

1. Technology adoption
2. Social movements
3. Cultural change
4. Political institutions
5. Insurance claims
6. Accounting and audit
7. Institutional reform
8. Personal identity development

A.3 The three strongest discovery signatures

Across all four groups, the most important empirical signatures are the following.

Signature 1 — Hidden progression under visible stability

$R(t) \approx$ constant while $Q(t)$ changes materially. (A.1)

Therefore:

$\theta(t) = \operatorname{atan2}[Q(t),R(t)]$ changes even though the visible state appears stable. (A.2)

This signature is especially relevant to:

- bank stress;
- physiological compensation;
- ecosystem fragility;
- cybersecurity intrusion;
- battery degradation;
- cultural tension.

Signature 2 — Equal internal distance, unequal calendar duration

$\Delta t_1 \neq \Delta t_2$ while $\Delta\tau_{i,1} \approx \Delta\tau_{i,2}$. (A.3)

This signature is especially relevant to:

- crises;
- legal cases;
- projects;
- learning;

- recovery;
- scientific development;
- epidemics;
- technology adoption.

Signature 3 — Gate concentration by phase

$\Pr[\text{Gate} = 1 \mid \theta] > \Pr[\text{Gate} = 1 \mid t]$. (A.4)

This signature is especially relevant to:

- default;
- decompensation;
- release;
- judgment;
- cell-fate commitment;
- outage;
- settlement;
- policy implementation.

A.4 Domain triage matrix

Before developing a full model, a domain may be screened using four questions.

Question	Weak answer	Strong answer
Does elapsed time misrepresent internal progress?	Rarely	Repeatedly and systematically
Is there an independently measurable Q?	Q is inferred after failure	Q is observable before gates
Does phase align episodes?	Case-specific alignment	Cross-episode compression
Do gates occur by phase?	No stable relation	Reproducible phase windows

A domain is a strong initial candidate when all four answers fall on the strong side.

Appendix B — Notation and Variable Dictionary

B.1 Core variables

Symbol	Meaning

Symbol	Meaning
X	Larger parent system or field
Σo	Undeclared possibility field
P	Declared protocol
B	Boundary
Δ	Observation or aggregation rule
h	Time or state horizon
u	Admissible intervention family
q	Baseline environment
φ	Feature map
R	Admitted, expressed, verified, or committed state
Q	Retained, latent, reactive, compensatory, or unresolved conjugate state
Z	Complex state
A	Total declared magnitude
θ	Complex phase
τ_i	Accumulated internal phase time
t	Parent-world calendar time
k	Ledger-event index
G	Event gate
L_k	Ledger after k committed events
r_k	Residual associated with event k
ε	Unmodelled forcing or error
ω	Angular velocity dθ/dt
g_A	Relative amplitude-growth rate Ȧ/A

B.2 Complex-state definitions

$Z(t) = R(t) + iQ(t)$. (B.1)

$Z(t) = A(t) \exp[i\theta(t)]$. (B.2)

$A^2(t) = R^2(t) + Q^2(t)$. (B.3)

$R(t) = A(t) \cos \theta(t)$. (B.4)

$Q(t) = A(t) \sin \theta(t)$. (B.5)

$\theta(t) = \text{atan2}[Q(t),R(t)]$. (B.6)

B.3 Dynamic definitions

$dZ/dt = [g_A(t) + i\omega(t)]Z(t) + \varepsilon(t)$. (B.7)

$g_A(t) = \dot{A}(t)/A(t)$. (B.8)

$\omega(t) = \dot{\theta}(t)$. (B.9)

Under slowly changing amplitude and bounded residual:

$dZ/d\theta \approx iZ$. (B.10)

$dR/d\theta \approx -Q$. (B.11)

$dQ/d\theta \approx R$. (B.12)

B.4 Internal-time definitions

For locally monotonic phase:

$\tau_i(t) = \text{unwrap}[\theta(t)]$. (B.13)

For cumulative phase distance:

$\tau_i(t) = \int_0^t |\theta(s)|\, ds$. (B.14)

Direction variable:

$o(t) = \text{sgn}[\theta(t)]$. (B.15)

Complete branch-aware internal-time state:

$T_i(t) = [\tau_i(t), o(t), b(t)]$. (B.16)

where $b(t)$ denotes the phase branch.

B.5 Gate and ledger definitions

A gate converts the phase-bearing state into a consequential event:

$e_k = \text{Gate_P}[Z(\tau_i), L_k]$. (B.17)

The ledger then updates:

$L_{k+1} = \text{U_L}[L_k, e_k, r_k]$. (B.18)

The parent-world intervention is:

$u_k = \text{Policy}[Z_k, L_k]$. (B.19)

The parent world responds:

$X_{k+1} = F[X_k, u_k, \xi_k]$. (B.20)

This gives the reflexive loop:

$X_k \rightarrow Z_k \rightarrow \text{Gate}_k \rightarrow L_{k+1} \rightarrow u_k \rightarrow X_{k+1}$. (B.21)

B.6 World-formation definitions

$P = (B, \Delta, h, u)$. (B.22)

$\text{World_P} = (X, q, \varphi, P)$. (B.23)

The complete proposed chain is:

Σ_0
→ Declare_P
→ Project_P
→ Z_P = R_P + iQ_P
→ θ_P
→ τ_i,P
→ Gate_P
→ Trace_P + Residual_P
→ InvarianceTest_P
→ Revise_P
→ Σ'. (B.24)

B.7 Important distinctions

Q versus residual

Q is part of the declared complex state.

Residual is what remains outside that model.

Q_P ≠ Residual_P. (B.25)

Phase versus time

Phase is orientation.

Internal time is accumulated or ordered phase progression.

$\theta \neq \tau_i$. (B.26)

Internal time versus ledger time

τ_i describes continuous internal progression.

k counts committed events.

$\tau_i \neq k$. (B.27)

Gate versus observation

Projection makes structure visible.

Gate makes a transition consequential.

Projection_P ≠ Gate_P. (B.28)

Appendix C — Complex-First Assessment Checklist

The following checklist is intended for researchers deciding whether a new domain should be investigated using a complex secondary-time framework.

C.1 Protocol declaration

Question	Yes / No
Is the system boundary B explicit?	
Is the observation rule Δ explicit?	
Is the analysis horizon h explicit?	
Are admissible interventions u explicit?	
Is the baseline q defined?	
Is the feature map φ defined?	
Are gate and trace rules declared?	
Is residual handling declared?	

A domain should not proceed to complexification if these elements are unstable or hidden.

C.2 Real-side coordinate R

Question	Yes / No
Does R have an independent operational definition?	
Can R be measured before the target event?	
Does R represent admitted, expressed, or verified structure?	
Is R comparable across episodes?	
Is measurement uncertainty recorded?	

C.3 Conjugate coordinate Q

Question	Yes / No
Is Q distinct from R?	
Can Q be estimated independently?	
Does Q influence later movement in R?	
Does Q support, oppose, defer, or convert into R?	
Is Q defined before the event gate?	
Does Q retain the same meaning across episodes?	
Is Q separate from generic residual error?	

A candidate Q should be rejected when most of these answers are negative.

C.4 Complex geometry

Question	Yes / No
Does A have an independent domain meaning?	
Is the normalization of R and Q defensible?	
Is θ robust to reasonable scaling changes?	
Does multiplication by i correspond to a meaningful transformation?	
Is phase more than a visual angle?	

Question	Yes / No
Does phase have an interpretable direction?	
Are branch reversals explicitly handled?	

C.5 Secondary-time test

Question	Yes / No
Do comparable episodes differ substantially in calendar duration?	
Do they become more similar after phase alignment?	
Is τ_i more stable than t as an internal progress measure?	
Does phase reduce model complexity?	
Does phase improve out-of-sample prediction?	
Does phase remain useful across regimes or observers?	

A minimal secondary-time claim requires strong evidence in this section.

C.6 Gate and ledger test

Question	Yes / No
Are consequential gates independently identifiable?	
Do gates occur in stable phase regions?	
Does phase outperform calendar time in gate prediction?	
Does each gate write persistent trace?	
Does the trace influence future action?	
Is residual attached to each closure?	
Is revision possible without erasing prior trace?	

C.7 Backreaction test

Question	Yes / No
Does the secondary-world output alter parent-world behaviour?	
Does the gate change resources, incentives, or constraints?	
Does the ledger alter future admissibility?	
Does phase-aware intervention improve outcomes?	
Is the resulting parent-world change measurable?	

The strongest world-forming claim requires positive answers here.

C.8 Reduction decision

After testing, choose the simplest sufficient model.

Use one real variable R when:

- Q adds no value;

- no meaningful conjugate channel exists;
- phase is arbitrary.

Use a real pair (R,Q) when:

- both variables are useful;
- no special complex geometry is needed;
- phase adds no simplification.

Use a complex state R + iQ when:

- R and Q form a conjugate pair;
- phase has operational meaning;
- complex structure simplifies dynamics.

Use secondary phase time when:

- phase aligns unequal-duration episodes;
- τ_i supplies stable internal order.

Use a phase–gate model when:

- consequential events are phase-sensitive.

Use a full secondary-world model when:

- gates write trace;
- trace changes future admissibility;
- outputs backreact on the parent world.

The reduction ladder is:

Full world
→ phase–gate model
→ phase-time model
→ complex state
→ real pair
→ real scalar. (C.1)

Appendix D — Evidence Ladder and Claim Language

D.1 Why claim language matters

A new cross-domain framework can lose credibility when its language advances faster than its evidence.

The following evidence ladder separates representation, dynamics, internal time, event formation, and world formation.

D.2 Evidence level E1 — Two-channel representation

Required evidence

- R and Q are distinct;
- both are measurable;
- each adds useful information.

Appropriate language

> The domain supports a two-channel state representation.

Avoid

> The domain has imaginary time.

D.3 Evidence level E2 — Complex completion

Required evidence

- A and θ have operational meaning;
- normalization is defensible;
- phase is robust;
- complex notation improves interpretation.

Appropriate language

> The domain admits a meaningful complex-state representation.

Avoid

> The domain is intrinsically complex in the physical sense.

D.4 Evidence level E3 — Phase-bearing dynamics

Required evidence

- phase evolves coherently;
- $dZ/d\theta$ is simpler or more stable than dZ/dt;
- phase improves prediction or classification.

Appropriate language

The domain exhibits phase-bearing dynamics.

Avoid

Phase has already become time.

D.5 Evidence level E4 — Local secondary time

Required evidence

- unequal-duration episodes align under τ_i;
- phase progression is locally monotonic or branch-controlled;
- internal order transfers across episodes.

Appropriate language

The domain supports a locally useful secondary phase time.

Avoid

The domain possesses a universal global clock.

D.6 Evidence level E5 — Phase-sensitive event formation

Required evidence

- consequential gates occur within reproducible phase regions;
- phase predicts gates better than calendar time;
- the result survives real-model comparison.

Appropriate language

The domain supports phase-sensitive event formation.

Avoid

Every event is caused by phase alone.

D.7 Evidence level E6 — Ledgered secondary world

Required evidence

- gates write persistent trace;
- trace affects future admissibility;
- observers can reconcile ledger order;
- backreaction alters the parent system;
- revision preserves prior trace and residual.

Appropriate language

The domain supports a ledgered secondary time-bearing world interpretation.

Avoid

A literal independent universe has been proven.

D.8 Evidence summary table

Level	Core achievement	Permissible claim
E1	Distinct R and Q	Two-channel representation
E2	Meaningful A and θ	Complex-state model
E3	Dynamically useful phase	Phase-bearing dynamics
E4	Unequal episodes align	Local secondary phase time
E5	Gates concentrate by phase	Phase-sensitive event formation
E6	Trace and backreaction form history	Ledgered secondary time-bearing world

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D.9 Recommended falsification statements

Every empirical paper applying the framework should state conditions under which its claim would fail.

Examples include:

The complex interpretation fails if Q cannot be independently estimated. (D.1)

The phase-time interpretation fails if $D_\theta \geq D_t$. (D.2)

The gate interpretation fails if phase does not improve event prediction. (D.3)

The complex privilege fails if a flexible real model performs equally well. (D.4)

The secondary-world interpretation fails if gates produce no persistent trace or backreaction. (D.5)

The global-clock interpretation fails if phase branches cannot be reconciled. (D.6)

D.10 Standard article footer for future domain studies

A future domain-specific study may conclude with the following audit block.

Declared protocol

$P = (B,\Delta,h,u)$. (D.7)

Real-side observable

R = admitted or verified state under P. (D.8)

Conjugate candidate

Q = independently measurable retained structure under P. (D.9)

Complex state

$Z = R + iQ = A \exp(i\theta)$. (D.10)

Internal time

τ_i = declared accumulated phase coordinate. (D.11)

Gate

G = declared consequential transition rule. (D.12)

Trace

$L_{k+1} = \text{UpdateLedger}(L_k, e_k, r_k)$. (D.13)

Classical comparison

Complex model compared against:

- calendar-time baseline;
- real scalar model;
- flexible real pair;
- standard domain model.

Residual disclosure

List:

- unexplained variance;
- branch ambiguity;
- failed episodes;
- observer disagreement;
- scaling sensitivity;
- omitted variables.

Reduction rule

State the simpler model to which the framework should return if the claimed gain disappears.

Closing Research Proposition

The complete research proposition can now be stated in its most disciplined form:

> Begin with domains in which elapsed duration fails to represent internal progress. Identify an admitted real-side state and an independently measurable conjugate channel. Test whether their complex phase supplies a simpler and more transferable internal ordering. Then determine whether consequential gates occur by phase, whether those gates write persistent trace, and whether the resulting ledger changes the future parent world.

In compact form:

$R + iQ$
$\rightarrow \theta$
$\rightarrow \tau_i$
$\rightarrow$ Gate
$\rightarrow$ Trace
$\rightarrow$ Backreaction
$\rightarrow$ Time-bearing world. (D.14)

The complex representation earns priority only when every arrow contributes measurable structure.

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.